

UGANDA NATIONAL EXAMINATIONS BOARD
PRIMARY LEAVING EXAMINATION

2025

INTEGRATED SCIENCE

Time Allowed: 2 hours 15 minutes

Total Marks scored	
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Random No.					Personal No.		

Candidate's Name:

Candidate's Signature.

District ID No.

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Read the following instructions carefully:

1. Do not write your **school** or **district name** anywhere on this paper.
2. This paper has two Sections: **A** and **B**.
Section **A** has **40** questions and Section **B** has 15 questions. The paper has **15 printed pages** altogether.
3. Answer **all** questions. **All** answers to both Sections **A** and **B** must be written in the spaces provided.
4. All answers **must** be written using a **blue** or **black** ballpoint pen or ink. Any work written in pencil, other than drawings, will **not** be marked.
5. Unnecessary **changes** in your work and handwriting that cannot be easily read may lead to **loss** of marks.
6. Do not fill anything in the table indicated **"FOR EXAMINERS' USE ONLY"** and in the boxes inside the question paper.

FOR EXAMINERS' USE ONLY		
QN. NO.	MARKS	EXR'S NO.
1 – 10		
11 – 20		
21 – 30		
31 – 40		
41 – 43		
44 – 46		
47 – 49		
50 – 52		
53 – 55		
TOTAL		

SECTION A: 40 MARKS

*Answer **all** questions in this section.
Questions **1** to **40** carry **one** mark each.*

1. Name the type of soil which is good for the growing of crops.

- ✓ Loam soil

2. Give the function of the stigma of a flower.

- ✓ The stigma receives pollen grains during pollination.

3. State any one way of controlling heartburn in humans.

- ✓ Avoid eating spicy or fatty or oily foods.
- ✓ Avoid overeating / Eat small amounts of food.
- ✓ Doing some light physical exercises after eating.

4. State any one way in which slashing of bushes around a home helps to control diseases.

- ✓ Slashing of bushes destroys hiding places for disease vectors.

5. Mention any one way in which planting of trees improves the quality of air in the environment.

- ✓ Trees purify air by absorbing carbon dioxide and releasing oxygen.
- ✓ Trees take in carbon dioxide from air and release oxygen.

6. Give any one way that can be used to prevent the spread of tuberculosis in a community.

- ✓ Taking children for immunisation.
- ✓ Isolating infected people/children.
- ✓ Drinking boiled milk.

7. State any one reason why it is not good to breathe through the mouth.

- ✓ There are no cilia to filter the air.
- ✓ The mouth does not filter the air.
- ✓ The mouth does not filter, warm or moisten the air.

The diagram below shows part of a root tuber crop. Use it to answer questions 8 and 9.



8. Name any one method of harvesting the root tuber crop.

- ✓ Uprooting / Digging out

9. Give any one reason why the above root tuber crop should be preserved after harvesting.

- ✓ To prevent it from rotting.
- ✓ To keep it for a longer time.
- ✓ To make it easy to transport.
- ✓ To promote food security.

10. State any one way through which breast feeding prevents diseases in babies.

- ✓ Breast milk provides antibodies that boost the immunity of the baby.

11. Give any one characteristic of bacteria.

- ✓ They are single celled.
- ✓ They have no nucleus.
- ✓ They reproduce by binary fission / cell division.
- ✓ They have no uniform shape and size.
- ✓ They are microscopic.
- ✓ They feed parasitically or saprophytically.

12. Mention the use of a barometer in a weather station.

- ✓ A barometer is used to measure the air pressure.
- ✓ A barometer is used to measure atmospheric pressure.

13. State any one way through which the giving of feeds to a cow during milking helps to stimulate milk let down.

- ✓ Keeps the cow calm

14. Give one way in which the wearing of tight-fitting shorts helps in controlling thread worm infestation.

- ✓ Prevents threadworms from laying eggs around the anus.

15. Name the part of the human male reproductive system that stores sperm cells.

- ✓ Epididymis

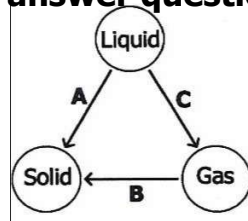
16. Give any one way of conserving living organisms in the soil.

- ✓ By mulching the soil
- ✓ Avoid disposing plastics on soil.
- ✓ By growing legumes.

17. State any one effect of weather on the pitch of sound.

- ✓ Pitch of sound increases with increasing temperature.
- ✓ Pitch of sound increases with increasing humidity.

The diagram below shows the physical changes that take place in the states of matter. Use it to answer questions 18 and 19.



18. Name the physical change labelled A.

- ✓ Freezing

19. State any one way in which physical change labelled C is important to plants.

- ✓ Change C helps plants to cool themselves.
- ✓ Helps in transpiration to cool the plants.

20. Apart from being a drink, give any one other use of alcohol.

- ✓ Used as a disinfectant.
- ✓ Used for cleaning wounds.
- ✓ Used for mixing drugs.
- ✓ Used for sterilizing medical equipment.

21. Name any one exotic breed of goats kept for the same purpose as Friesian cows.

- ✓ Alpine goats
- ✓ Anglo-Nubian goats
- ✓ Toggenburg goats
- ✓ Saanen goats

22. State any one way in which flowers planted in a school compound are important.

- ✓ Flowers beautify the school compound.
- ✓ Flowers are used for decoration.
- ✓ Some flowers are eaten as food.
- ✓ Flowers are used for study research / purposes.

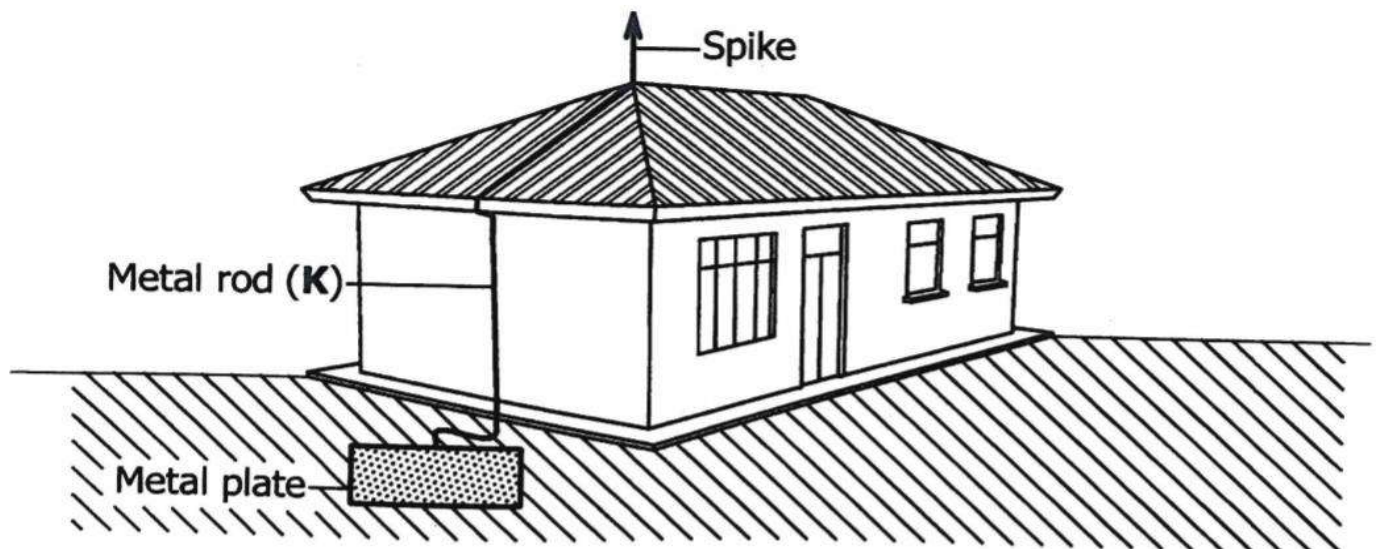
23. Give any one way of keeping the human body muscles in good working condition.

- ✓ Doing regular body / physical exercises.
- ✓ Feeding on a balanced diet.

24. Mention any one method used to prepare food at home.

- | | |
|------------|------------|
| ✓ Frying | ✓ Steaming |
| ✓ Boiling | ✓ Mingling |
| ✓ Roasting | ✓ Baking |

The diagram below shows a lightning conductor on a building. Use it to answer questions 25 and 26.



25. Name the material used to make the metal rod labelled K.

- ✓ Copper

26. State one way in which the above lightning conductor helps to control damage on buildings.

- ✓ Lightning conductor traps static charges and directs them safely to the ground.
- ✓ Lightning conductor conducts excess electric charges from the clouds into the ground.

27. Give one reason why boiled water should be filtered before drinking.

- ✓ To remove solid impurities from water

28. Mention any one way in which wind is used as a resource.

- ✓ Wind helps in winnowing harvested cereals
- ✓ Wind turns windmills to generate wind power.
- ✓ Wind helps in sailing boats.
- ✓ Wind helps in cross pollination.
- ✓ Wind helps in seed dispersal.

29. State any one way in which the strength of a magnet can be maintained.

- ✓ By keeping magnets with iron keepers.
- ✓ By painting magnets to prevent rusting.
- ✓ By keeping magnets with unlike poles together.
- ✓ By keeping magnets away from heat.
- ✓ By keeping magnets facing the North-South direction.

30. Give any one factor a health worker considers when prescribing drugs for a patient.

- ✓ Age of the patient
- ✓ Weight of the patient
- ✓ Type of disease

31. Identify the component of blood that transports waste products in the human body.

- ✓ Plasma

32. Apart from being less dense, state any one other factor that makes an object float on a liquid.

- ✓ The shape of the object
- ✓ Upthrust force (Buoyancy)

33. State any one way in which animals benefit from moving in groups.

- ✓ Animals get protection from predators/enemies
- ✓ Animals can easily find mates
- ✓ Animals can easily find food and water

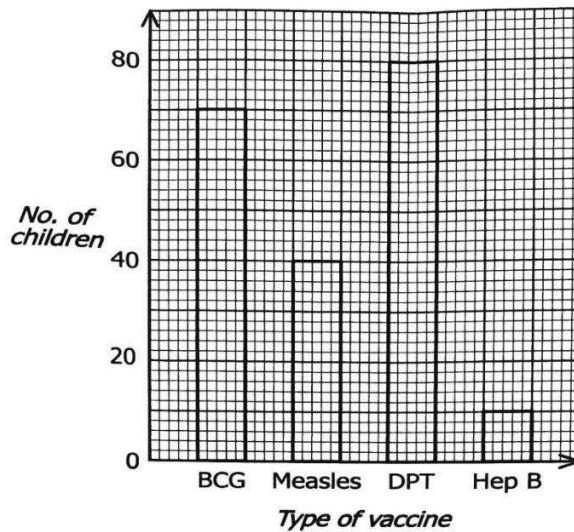
34. Name any one feature you can use to identify combs with young bees in a hive.

- ✓ Presence of brood capped cells.
- ✓ They are dull coloured.

35. State the reason why sound travels fastest in solids.

- ✓ Particles in solids are very closely packed together than in liquids and gases which makes heat conduction through solids fastest.
- ✓ Particles in solids are very closely packed together allowing atomic vibrations to conduct heat more faster than in liquids and gases.

The graph below shows vaccines that were received by children at a health Centre on one Tuesday. Use it to answer questions 36 and 37.



36. Identify any one disease for which the highest number of children received the vaccine.

- ✓ Diphtheria
- ✓ Pertussis / whooping cough
- ✓ Tetanus

37. Determine the number of children born on that Tuesday.

- ✓ Seventy (70) children were born on that Tuesday.

38. State the reason why a balloon filled with air bursts when exposed to sunshine.

- ✓ Due to rapid expansion of air inside the balloon caused by heating.

39. Give any one practice of caring for the health of the brain in humans.

- ✓ Have enough rest and sleep to refresh the brain.
- ✓ Feed on a balanced diet.
- ✓ Do regular physical exercises.
- ✓ Avoid tobacco smoking and alcoholism.
- ✓ Immunise children against meningitis.

40. State any one way through which establishment of factories in a community can be dangerous to water sources.

- ✓ Factories release untreated wastes into water bodies.
- ✓ Establishment of factories leads to destruction (draining away) of water sources.

SECTION B: 60 MARKS

Answer all questions in this section.

Questions 41 to 55 carry four marks each.

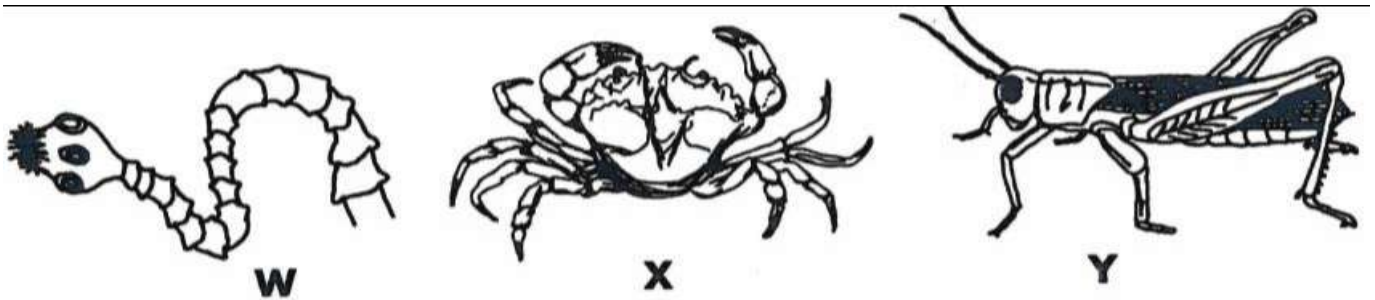
41. (a) Give any two social changes that take place at puberty.

- ✓ Adolescents become interested in opposite sex.
- ✓ Adolescents move in peer groups.
- ✓ Adolescents develop desire for freedom.
- ✓ Adolescents rebel against rules of teachers/parents.
- ✓ Adolescents become disobedient to their parents/teachers.

(b) Mention any two ways through which an adolescent can manage the social changes.

- ✓ Joining good social clubs.
- ✓ Joining good peer groups / peer resistance.
- ✓ Through making good decisions (good decision making).
- ✓ Through assertiveness.

42. The diagrams below show the animals W, X and Y that belong to three different groups of invertebrates. Use them to answer questions that follow.



(a) Identify the group of invertebrates to which animal labelled X belongs.

- ✓ Crustaceans

(b) Apart from having no backbone, state any one other way animals W and Y are similar in their body structures.

- ✓ Animal X and Y have segmented bodies
- ✓ Their bodies are both covered with a protective covering.
- ✓ Both have their bodies divided into two equal halves along one plane (Bilateral Symmetry).

(c) State any two benefits of animal Y to humans.

- ✓ Animal Y is eaten as food.
- ✓ Animal Y is sold for money.
- ✓ Animal Y pollinates farmer's crops.

43. (a) Give the importance of each of the following in the human diet:

(i) Roughage.

- ✓ Prevents constipation.
- ✓ Adds bulk to the diet.
- ✓ Allows easy digestion of food.

(ii) Water.

- ✓ Aids food digestion.

(b) Give any two ways in which food taboos are important to people in some communities.

- ✓ They prevent people from eating harmful food.
- ✓ They preserve culture and identity.
- ✓ They conserve natural resources.
- ✓ They promote respect and discipline.
- ✓ They shape people's habits.

44. In the table below, A shows the elements of Primary Health Care (PHC) while B shows some of the problems that may result from failure to practice the elements of PHC.

A	B
Immunisation	Typhoid fever
Family planning	Constipation
Water and sanitation	Halitosis
Oral health	Hepatitis B
	Maternal anaemia

Match correctly the problems in B with the elements of PHC in A, in spaces provided below.

- (a) **Immunisation** – Hepatitis B
- (b) **Family planning** – Maternal anaemia
- (c) **Water and Sanitation** – Typhoid fever
- (d) **Oral health** – Halitosis

45. A farmer has an apiary, a goat unit and a coffee plantation on the same piece of land.

(a) State the benefit of the;

(i) apiary to the coffee plantation.

- ✓ Bees pollinate coffee plants.

(ii) a goat unit to the coffee plantation.

- ✓ Droppings from goats is used as fertilizer in coffee plantation

(b) Mention any one danger of the apiary to the goat unit.

- ✓ By tethering goats
- ✓ By keeping goats indoors
- ✓ Carrying out Zero grazing

(d) Give any one way in which the farmer can control goats from disturbing the apiary.

- ✓ Fencing off the goat unit from the apiary

46. The diagrams below show experiments that were carried out on how light travels. Use them to answer the questions that follow.



(a) Identify the experiment in which the observer was able to see the light.

- ✓ Experiment N

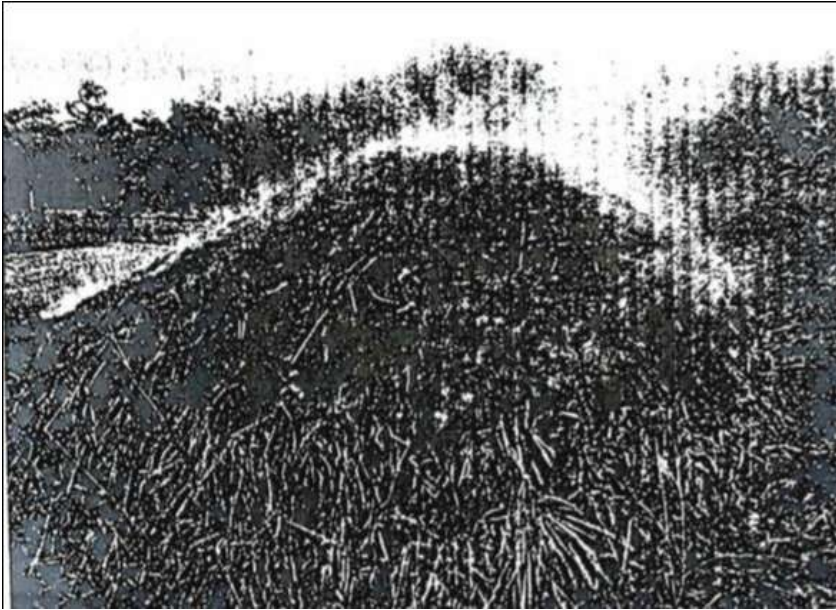
(b) State the principle of light demonstrated in the experiments above.

- ✓ Light travels in a straight line.

(c) Explain how the principle in (b) above results in the formation of shadows. (02 marks)

- ✓ When light rays travelling in a straight line meet an opaque object, they are obstructed and since they cannot bend over the opaque object, a shadow is formed on the opposite side of the opaque object.

47. The diagram below shows part of the process of charcoal making. Use it to answer the questions that follow.



(a) State the reason for covering the wood with soil in the above process.

- ✓ To limit the supply of oxygen to the burning wood.

(b) State any one danger of the above activity to the environment.

- ✓ Produces smoke that pollutes air.
- ✓ Leads to deforestation.

(c) Identify any two ways the above activity can be controlled in a community.

- ✓ Enforcing laws against deforestation.
- ✓ Using alternative sources of energy.
- ✓ Using electricity for cooking instead of charcoal.

48. (a) State any two reasons why rubbish from our homes should be sorted before disposing it to gardens.

- ✓ The rubbish can pollute the soil.
- ✓ Rubbish can kill soil organisms.
- ✓ Rubbish can lead to waterlogging if not sorted.
- ✓ Rubbish can damage soil structure and texture.
- ✓ Rubbish can block water drainage in the soil.
- ✓ Rubbish can block soil aeration.

(b) Mention any two ways of keeping yourself safe while sorting rubbish at home.

- ✓ Wearing hard rubber gloves before sorting rubbish.
- ✓ Wearing gumboots to protect the feet from germs.
- ✓ Wearing a face mask to prevent direct inhaling of contaminated air.

49. (a) State any two ways you can use to control common sicknesses at home.

- ✓ Feeding on a balanced diet.
- ✓ Boiling water for drinking.
- ✓ Promoting proper personal hygiene.
- ✓ Proper disposal of wastes.
- ✓ Having regular body exercises.
- ✓ Promoting proper food hygiene.
- ✓ Avoid alcoholism, smoking and drug abuse.

(b) Give any two ways through which juvenile delinquency can be controlled in a society.

- ✓ Educating children.
- ✓ Punishing children for wrong behavior.
- ✓ Providing guidance and counselling to children.
- ✓ Providing equal treatment to all children.

50. (a) Name one material excreted by the lungs in humans.

- ✓ Carbon dioxide / Water vapour

(b) State one way through which shivering helps to regulate body temperature.

- ✓ Muscles contract rapidly to produce heat.

(c) Explain how each of the following practices helps in caring for the skin:

(i) Wearing clean loose clothes

- ✓ They keep the skin free from germs.
- ✓ They prevent skin irritation.
- ✓ They prevent bad odour/smell.

(ii) Smearing with Vaseline.

- ✓ Vaseline keeps the skin soft and moist.

51. (a) State any two uses of the sun as an energy resource.

- ✓ Sunlight is used in photography.
- ✓ Heat from the sun is used to produce solar electricity.
- ✓ Heat from the sun is used to dry wet clothes.
- ✓ Heat from the sun is used to dry harvested crops.
- ✓ The sun provides light that enables us to see.

(b) State any one reason why fire extinguishers are commonly painted red.

- ✓ To be easily seen in case fire breaks out.
- ✓ For easy identification in case of fire outbreak.

(c) State one way in which soil is useful in putting out fire.

- ✓ Soil cuts off the supply of oxygen to stop the burning.

52. (a) Give any one example of a good conductor of heat.

- | | |
|----------|------------|
| ✓ Iron | ✓ Lead |
| ✓ Copper | ✓ Zinc |
| ✓ Silver | ✓ Mercury |
| ✓ Brass | ✓ Aluminum |

(b) Mention any one way in which heat transfer by radiation is important to;

(i) plants.

- ✓ Helps plants to transpire.
- ✓ Helps in seed germination.

(ii) humans.

- ✓ Helps in drying harvested crops.
- ✓ Helps in drying people's clothes/beddings.
- ✓ Enables people to warm themselves.

(c) Explain how the silvery walls of a thermos flask keep the hot liquid in the flask hot.

- ✓ They are polished to reflect radiant heat back into the liquid.

53. (a) State the reason why a person suffering from HIV/AIDS is easily attacked by other diseases.

- ✓ HIV/AIDS weakens the immunity of a person.
- ✓ HIV/AIDS destroys white blood cells which fight against germs in the body.

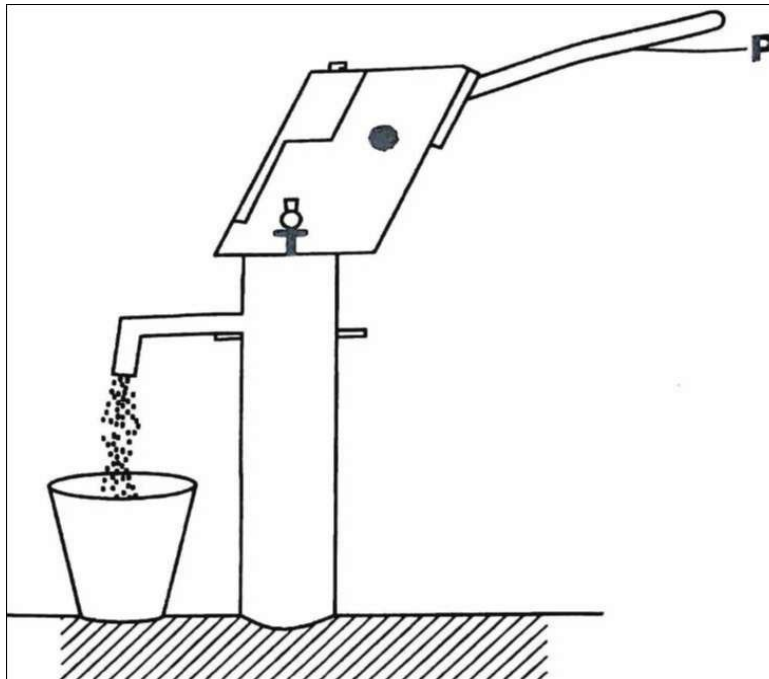
(b) Mention any one effect of HIV/AIDS on a community.

- ✓ It leads to loss of friends.
- ✓ It leads to loss of skilled labour in the community.
- ✓ It increases the number of street children.
- ✓ It leads to shortage of productive population.

(c) Give any two ways of caring for a person living with HIV/AIDS.

- ✓ Washing for them clothes.
- ✓ Feeding them on a balanced diet.
- ✓ Cooking for them food.
- ✓ Taking them to the hospital for treatment.
- ✓ Providing them guidance and counselling.

54. The diagram below is of a simple machine. Use it to answer the questions that follow.



(a) Name the class of levers to which the above machine belongs.

- ✓ First class levers

(b) Using an arrow, show the direction of the load when effort is applied to the part labelled P.

Note:

[Diagram of a first-class lever: fulcrum in the middle, effort on one side, load on the other, with arrow pointing to the load side labeled "Direction of the load"]

(c) State any two ways through which the efficiency of the above simple machine can be increased.

- ✓ Oiling the moving parts of the machine
- ✓ Greasing the moving parts of the machine
- ✓ Repairing the damaged parts of the machine

55. (a) Give any one cause of burns at home.

- ✓ Skin contact with acid
- ✓ Skin contact with hot ash
- ✓ Skin contact with hot charcoal
- ✓ Skin contact with hot metals

(b) Mention any one way in which cuts can be prevented in a school.

- ✓ Keeping sharp tools in safe places
- ✓ Using hard materials hand gloves to pick up broken sharp objects
- ✓ Throwing/disposing broken glasses into deep pits

(c) Mention any two responsibilities of a good first aider.

- ✓ To monitor the condition of the casualty
- ✓ To ensure safety of the casualty
- ✓ To provide appropriate first aid treatment
- ✓ To assess the situation quickly and calmly
- ✓ To call for medical help if needed.

END



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41 - 43		
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TOTAL		

SECTION A: 40 MARKS

*Answer all questions in this section.
Questions 1 to 40 carry one mark each.*

1. Name any **one** type of latrine.

.....

2. Give any **one** reason why it is dangerous to take shelter under tall trees during rain.

.....

.....

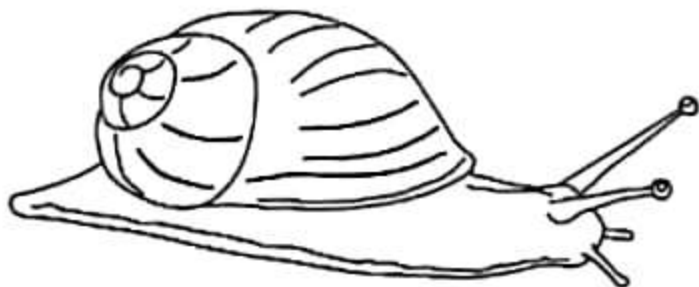
3. State any **one** function of the diaphragm during the process of breathing.

.....

4. Name any **one** agent of soil erosion.

.....

The diagram below shows an animal in the group of invertebrates. Use it to answer questions 5 and 6.



5. Name the animal shown in the diagram.

.....

6. State the group of invertebrates to which the animal belongs.

.....

7. Mention any **one** way in which vegetables can be kept fresh for some time.

.....

8. State **one** way in which bacteria are useful in the formation of soil.

.....

.....

9. Give the reason why a person suffering from bilharzia develops anaemia.

.....

.....

10. Mention **one** way in which air circulation in a rabbit hutch can be improved.

.....



11. State the reason why the human skin develops goose pimples during cold weather.

.....

.....

12. Give any **one** way in which you can utilise empty plastic bottles at home.

.....

13. Mention **one** function of a bandage found in a first aid kit.

.....

14. State any **one** way in which fermentation is important in our daily life.

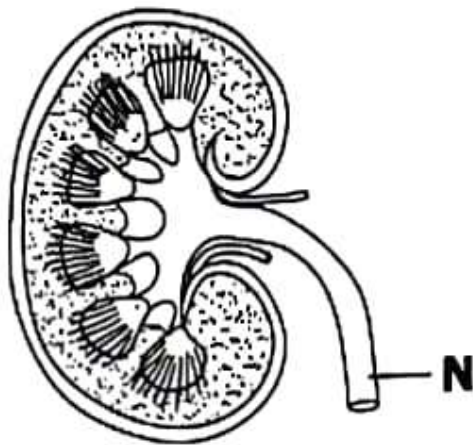
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15. State any **one** reason why a builder would prefer a pulley to an inclined plane when lifting sand from the ground to the top of a building.

.....

.....

The diagram below shows a human body organ. Use it to answer questions 16 and 17.



- 16. Name the organ shown above.
.....
- 17. Give the function of the part labelled N.
.....
- 18. Identify any **one** effect of pests and diseases on tuber crops.
.....
- 19. State any **one** way in which taking medicine without the instruction of a health worker can affect a family.
.....
- 20. Mention any **one** reason why harvesting of honey should be done during day time.
.....
.....
- 21. Name the disorder in the skeletal system where a bone moves away from its normal position at the joint.
.....

22. State the reason why a cubic unit of wool weighs less than a cubic unit of a stone.

.....

.....

23. State **one** way in which paddocking method of grazing cattle helps to break the life cycle of pests on a farm.

.....

24. Give **one** way in which rusting is important to the environment.

.....

25. Name the element of Primary Health Care which promotes the human body cleanliness.

.....

26. Mention any **one** activity a School Health Club can do to improve the health of people in school.

.....

27. State the property of air used in filling balloons.

.....

28. State any **one** way in which planting of trees alongside other crops improves the quality of soil.

.....

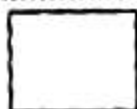
29. Give **one** reason why we should regularly wash our hands with clean water and soap.

.....

.....

30. Mention **one** way in which the antennae of insects are similar to the human skin in terms of function.

.....

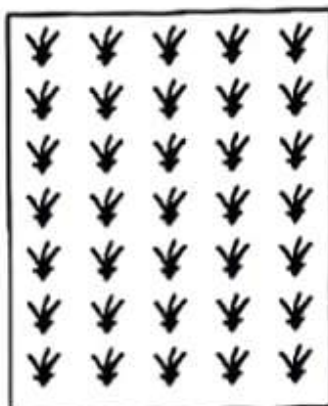


31. Give **one** reason why the vacuum in a thermos flask prevents heat loss by conduction and convection.

.....

.....

The diagram below shows a method of planting crops. Use it to answer questions 32 and 33.



32. Identify the method of planting crops shown above.

.....

33. State any **one** advantage of the method of planting crops shown in the diagram above.

.....

.....

34. Give any **one** example of an emotional change in adolescents.

.....

35. Name the immunisable viral disease which is spread due to poor sanitation.

.....

36. State the reason why a ray of light bends as it passes through a glass block.

.....

.....

37. State any **one** way of conserving water as a resource in the environment.

.....

38. Mention **one** reason why you are encouraged to brush the tongue as well when cleaning the teeth.

.....

.....

39. Mention any **one** way in which electric shocks can be prevented in a home.

.....

.....

40. Give any **one** way in which crop farmers can keep seeds viable for a long time.

.....

.....



SECTION B: 60 MARKS

*Answer all questions in this section.
Questions 41 to 55 carry four marks each.*

41. (a) Name the class of food that helps in the repair of body cells.
.....
- (b) Apart from contamination of food, mention any **two** other challenges that people who buy food from markets get.
(i)
(ii)
- (c) State any **one** way of controlling food contamination in a market.
.....
42. (a) Name the germ that causes cholera.
.....
- (b) Explain why cholera is common in heavily populated communities.
(02 marks)
.....
.....
.....
- (c) Apart from proper use of latrines, mention any **one** other way of controlling the spread of cholera in the communities in (b) above.
.....
43. (a) Name the correct garden tool for transplanting seedlings.
.....
- (b) State the reason why evening time is the best for transplanting seedlings.
.....
.....

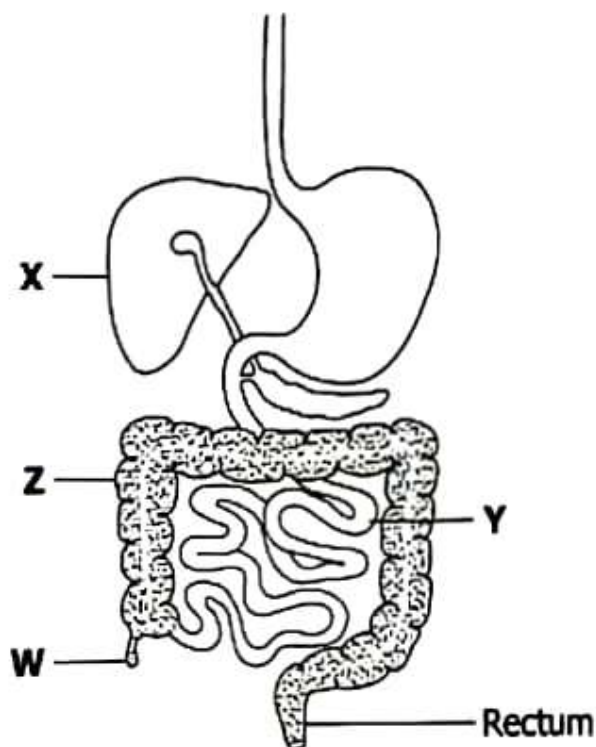
(c) Mention any **two** advantages of mulching a garden after transplanting.

(i)

(ii)



44. The diagram below shows part of the human digestive system. Use it to answer the questions that follow.



(a) Name the parts labelled;

(i) **X**

(ii) **W**

(b) State any **one** function of the following parts:

(i) **Y**

(ii) **Z**

45. The table below shows some of the steps followed in the cleaning of clothes and the purpose of each step. Complete the table correctly.

Step	Purpose
.....	Separating cloths according to washing needs.
Soaking	
.....	Removing dirty soapy water from the clothes.
Ironing	

46. (a) Give the meaning of the term **electromagnet**.

.....

- (b) You are provided with the following items: Iron nail, copper wire, a dry cell and some pins.

Describe how you can use the items to make a magnet.

(03 marks)

.....



47. (a) Mention any **two** materials that can be used in a digester to produce biogas.

(i)

(ii)

(b) Give a reason why a biogas digester should be tightly sealed.

.....

.....

(c) State any **one** advantage of using biogas.

.....

.....

48. The diagram below shows a person wearing a protective item made of cloth. Use it to answer the following questions.



(a) Name the protective item labelled K.

.....

(b) Identify any **two** diseases that can be prevented by the use of the protective item K.

(i)

(ii)

(c) Mention any **one** good practice of using the protective item K.

.....

49. (a) Mention **one** example of musical instruments in each of the groups below:

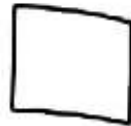
(i) String Instruments.....

(ii) Percussion Instruments.....

(b) Name any **two** methods of storing sound.

(i)

(ii)



50. (a) State the name given to Insects that live and work together.

.....

(b) Give any **one** way in which insects are useful to people.

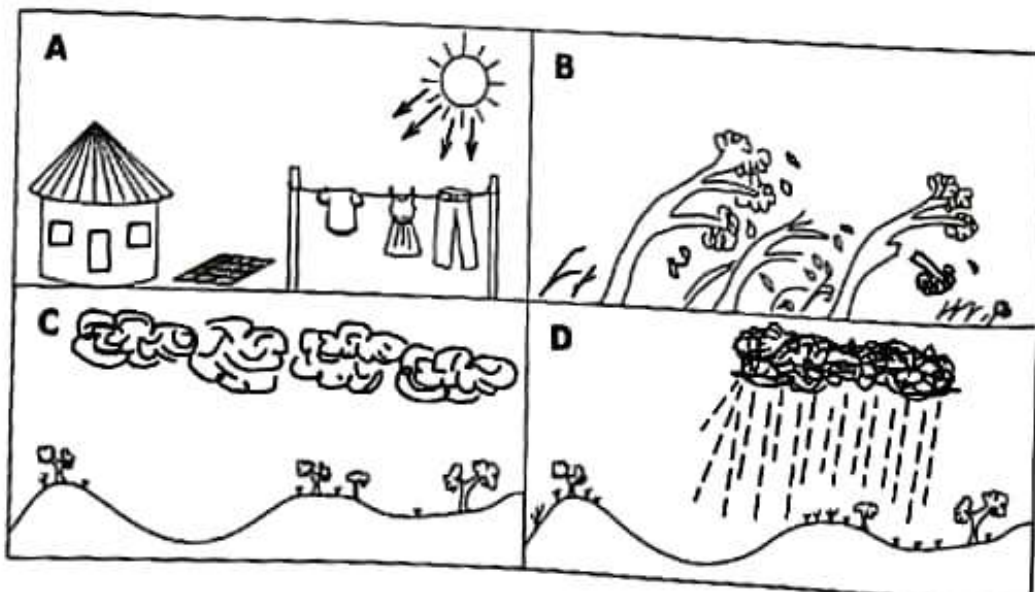
.....

(c) State any **two** human activities that are dangerous to insects in the environment.

(i)

(ii)

51. The diagram below shows types of weather. Study it and answer the questions that follow.



(a) Name the type of weather marked **C**.

.....

(b) State **one** way in which the type of weather in **A** helps in formation of the type of weather in **D**.

.....

(c) Give **one** way in which people protect themselves from the effects of the type of weather in **B**.

.....

(d) State any **one** way in which too much of the type of weather in **A** is dangerous to the environment.

.....

52. (a) State the meaning of the term **vaccine**.

.....

.....

(b) Mention the way in which polio vaccine is administered.

.....

(c) State the reason why vaccines should be stored in a cool place.

.....

.....

(d) Identify any **one** way you can support immunization programmes in your community.

.....



53. (a) Name the physical change which leads to the formation of the following:

(i) Ice

(ii) Rainfall

(b) State any **two** effects of physical changes caused by soil erosion in the environment.

(i)

(ii)

54. (a) Name the **two** muscles found in the human arm.

(i)

(ii)

(b) State any **one** function of muscles in the human body.

.....

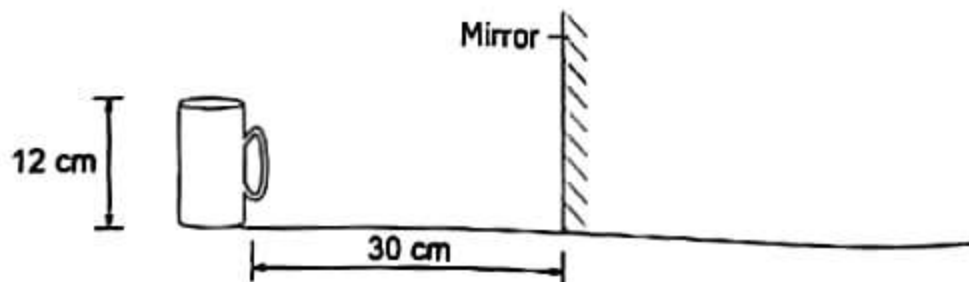
.....

(c) State the importance of maintaining correct posture in humans.

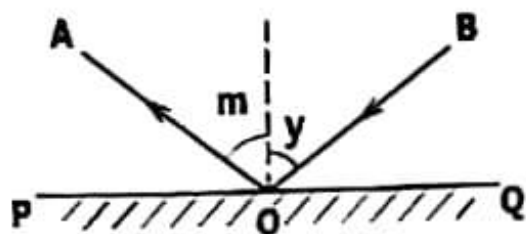
.....

.....

55. (a) The diagram below shows a cup 12 cm high placed 30 cm in front of a plane mirror. Draw the image of the cup. (02 marks)



- (b) The diagram below shows a reflection on a plane mirror **PQ**.



- (i) Name ray A.

.....

- (ii) State the relationship between angles **m** and **y**.

.....





UGANDA NATIONAL EXAMINATIONS BOARD
PRIMARY LEAVING EXAMINATION
2023

INTEGRATED SCIENCE

Time Allowed: 2 hours 15 minutes

Random No.						Personal No.		

Candidate's Name:

Candidate's Signature:

District ID No.

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Read the following instructions carefully:

1. Do not write your **school** or **district name** anywhere on this paper.
2. This paper has two sections: **A** and **B**. Section **A** has **40** questions and section **B** has **15** questions. The paper has **15 printed pages** altogether.
3. Answer **all** questions. **All** answers to both sections **A** and **B** must be written in the spaces provided.
4. All answers **must** be written using a **blue** or **black** ball point pen or ink. Any answer or work written in pencil other than drawings will **not** be marked.
5. Unnecessary **changes** in your work and handwriting that cannot be read easily may lead to **loss** of marks.
6. Do not fill anything in the table indicated: **"FOR EXAMINERS' USE ONLY"** and boxes inside the question paper.

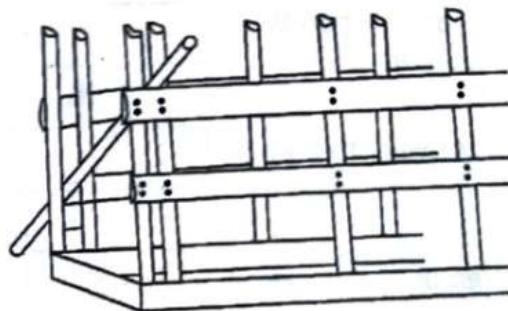
FOR EXAMINERS' USE ONLY		
QN. NO.	MARKS	EXR'S NO.
1 - 10		
11 - 20		
21 - 30		
31 - 40		
41 - 43		
44 - 46		
47 - 49		
50 - 52		
53 - 55		
TOTAL		

SECTION A: 40 MARKS

Questions 1 to 40 carry one mark each.

1. Name the type of teeth which develops last in human beings.
.....
2. Give any **one** way in which timely weeding of crops helps to control crop diseases.
.....
3. Give **one** element of weather which helps plants to make their own food.
.....
4. State the purpose of pressing the belly of a casualty of near drowning.
.....
5. Give any **one** reason why fish are able to live in water.
.....

The diagram below shows a structure in a cattle farm. Study and use it to answer questions 6 and 7.



6. Mention any **one** farm activity carried out in the above structure.
.....
7. How is the above structure useful when carrying out the farm activity you have mentioned above?
.....

8. Name the excretory organ that removes water and carbon dioxide from the human body.
-
9. Mention any **one** similarity between fungi and animals in the way they feed.
-
10. State **one** way in which the screen on the pipe of a VIP latrine helps to control diarrhoeal diseases.
-
11. State any **one** piece of advice you can give to children to help them prevent indigestion.
-
12. State any **one** way in which you can safeguard yourself against smoking.
-
13. Give the reason why most cooking pans are made of aluminium.
-
-
14. State **one** reason why the use of soap makes washing clothes easy.
-
-
15. Apart from deforestation, mention any **one** other human activity that can result into flooding in our environment.
-
16. Mention any **one** sign of roundworm infestation in humans.
-



The diagram below shows a metallic charcoal stove. Study and use it to answer questions **17** and **18**.



17. Give the function of the part labelled **M**.
.....
18. State **one** advantage of building the inner wall of the charcoal stove with clay.
.....
19. State any **one** disadvantage of people living with domestic animals in the same house.
.....
20. Give **one** way in which the use of biogas is helpful in the conservation of environment.
.....
21. State the effect of opaque objects on a beam of light.
.....
22. Apart from making candles, mention any **one** other way in which bees-wax is useful to people.
.....

23. State any **one** reason why farmers are encouraged to preserve food stuffs like sweet potatoes.

.....

.....

24. Name any **one** force which a pulley overcomes when it is being used.

.....

25. State **one** way in which a mask is important in maintaining the proper functioning of the respiratory system.

.....

.....

26. Apart from having bright colours, state any **one** other thing that attracts pollinators to flowers.

.....

27. State any **one** way of caring for simple garden tools.

.....

28. State **one** way in which the Child Health Card is important to a parent.

.....

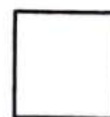
.....

29. State any **one** factor that affects the pitch of sound produced by an object.

.....

30. Name any **one** body part that may be checked during a school health parade.

.....



31. State any **one** way in which screws can be important in our daily life.

.....

.....

32. Apart from being a source of food, give the reason why farmers hang green vegetables in poultry houses.

.....

.....

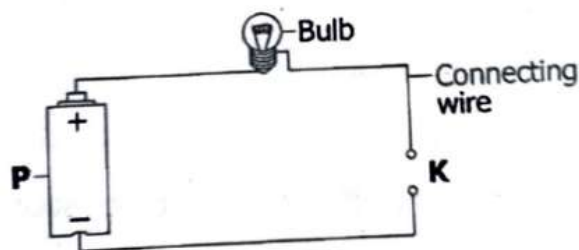
33. State any **one** change in food which shows that food has gone bad.

.....

34. State what would be observed when a plastic pen rubbed in hair is brought close to small pieces of paper.

.....

The diagram below represents a simple electric circuit. Study and use it to answer questions **35** to **36**.



35. Give the function of **P** in the circuit.

.....

36. Why would the bulb **not** light when a piece of glass is used to close

.....

.....

37. Name any **one** part of the human body where a gliding joint is found

.....

38. State any **one** reason why pruning of trees is important in agroforestry.

.....

.....

39. Write down any **one** activity that a School Environment Club can do to protect a school water source.

.....

.....

40. State **one** reason why a person suffering from HIV/AIDS is easily attacked by many other diseases.

.....

.....



SECTION B: 60 MARKS

Questions **41** to **55** carry four marks each.

41. (a) State any **two** effects of rusting on iron sheets.

(i)

(ii)

- (b) Identify **two** ways in which people protect iron sheets from rusting.

(i)

(ii)

42. The table below shows an example of a farm record. Study and use it to answer the questions that follow.

Date	Health problem	Animal(s) affected	Treatment/control method	Cost of treatment
01/8/2023	Mastitis	Goats No. 4 and No. 8	Antibiotics	17,000/=
12/10/2023	Worm infestation	Kids (06)	Dewormers	25,000/=
03/11/2023	Anthrax	All goats	Vaccination	50,000/=

- (a) Name the farm record above.

.....

- (b) Apart from Anthrax, mention any **one** other disease which is controlled by the method used on 03/11/2023.

.....

- (c) Give any **two** reasons why keeping of such a record is important to farmers.

(i)

(ii)

43. (a) Identify the type of soil erosion common in hilly areas.

.....

- (b) State any **one** danger of soil erosion to the environment.

.....

- (c) Explain how deforestation causes soil erosion. (02 marks)

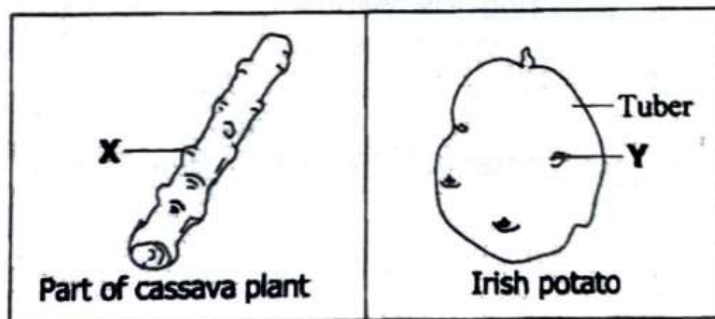
.....

.....

.....



44. The diagrams below show part of a cassava plant and an Irish potato that are used in propagation. Study and use them to answer the questions that follow.



- (a) Name the part of the cassava plant marked **X**.

.....

- (b) State the method of vegetative propagation where the part of a cassava plant shown above is used.

.....

- (c) In which **one** way is the function of part **X** of the cassava plant similar to that of part **Y** of the Irish potato?

.....

- (d) Give the importance of the tuber in the propagation of the Irish potato.

- 45 (a) What is meant by the term vulnerable group of people?

- (b) Apart from the sick, name **one** other group of vulnerable people in a community.

- (c) State any **two** ways in which you can care for the sick in your community.
 (i)
 (ii)
46. Moses wanted to find the volume of an irregular object. He had the following materials for the experiment; an irregular object, thread, overflow can and water.
- (a) Name **one** material Moses forgot to include on the list of his materials.

- (b) What is the use of the thread in the experiment?

- (c) Describe how Moses will get the volume of the irregular object in this experiment. (02 marks)



47. (a) Identify any **one** body change in girls during adolescence.

.....

- (b) State any **two** ways in which school boys can help to prevent teenage pregnancies.

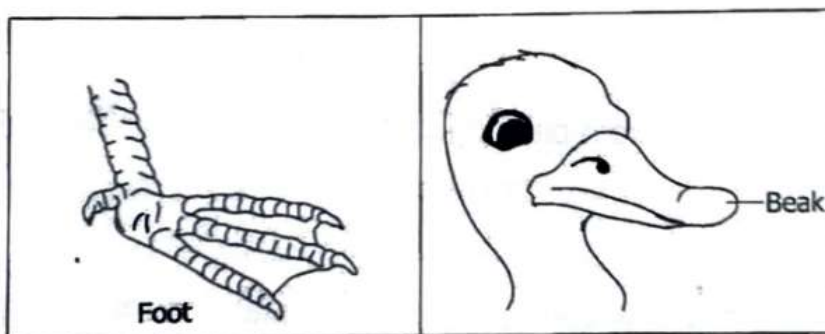
(i)

(ii)

- (c) State any **one** way of caring for the reproductive system in humans.

.....

48. The diagrams below are of a foot and a beak of a bird. Study and use them to answer the questions that follow.



- (a) Identify the group of birds with the type of foot and beak shown.

.....

- (b) How do the parts of the bird shown above help it live well in its environment?

(i) Foot:

.....

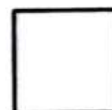
(ii) Beak:

.....

- (c) Give any **one** example of a domestic bird with such type of beak and foot.

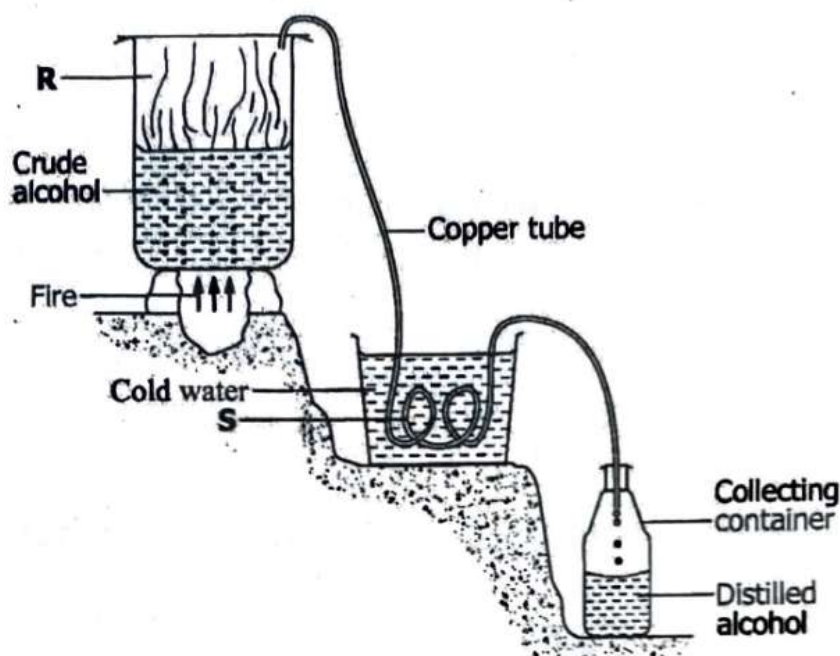
.....

49. (a) Give any **one** way in which each of the following elements of Primary Health Care (PHC) is important in a community.
- (i) Food and nutrition:
-
- (ii) Maternal and child health:
-
- (b) Identify any **two** ways of giving health education to a community.
- (i)
- (ii)



50. (a) Apart from planting crops in rows, give another method of planting crops.
-
- (b) Give any **two** advantages of planting maize in rows.
- (i)
- (ii)
- (c) State any **one** way in which the use of chemicals on a farm can be dangerous to a crop farmer.
-
-

51. The diagram below shows the distillation method of making alcohol. Study and use it to answer the questions that follow.



- (a) Identify the physical changes taking place;

(i) at part marked **R**.

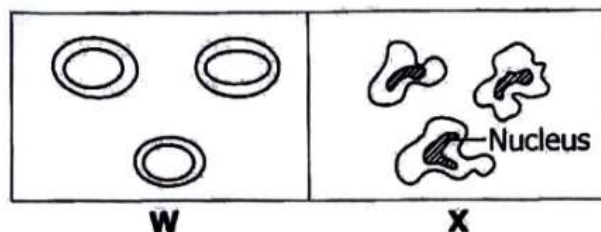
(ii) in part **S**.

- (b) Give **one** reason why the container with crude alcohol should be properly sealed.

- (c) Give the function of the copper tube in the method shown above.

52. (a) State any **two** health problems that are common in overpopulated areas.
- (i)
- (ii)
- (b) Mention any **two** ways in which collecting information on human population helps to address health problems in a community.
- (i)
- (ii)
53. (a) Give any **two** ways in which plants support human life.
- (i)
- (ii)
- (b) Mention any **two** ways in which humans can conserve plant life.
- (i)
- (ii)
54. (a) Identify any **two** characteristics of the female anopheles mosquito that makes it a good vector for malaria.
- (i)
- (ii)
- (b) How do the following methods help in controlling the spread of malaria?
- (i) Pouring oil on stagnant water.
-
-
- (ii) Sleeping under a mosquito net.
-
-

55. The diagrams below show the structure of different blood cells in the human body. Study and use them to answer the questions that follow.



- (a) Name the blood cell marked **X**.

.....

- (b) Give the function of the blood cells marked:

(i) **W**

(ii) **X**

- (c) Identify any **one** disease which destroys blood cells **W**.

.....





UGANDA NATIONAL EXAMINATIONS BOARD

PRIMARY LEAVING EXAMINATION

2022

INTEGRATED SCIENCE

Time Allowed: 2 hours 15 minutes

Random No.						Personal No.		

Candidate's Name:

Candidate's Signature:

District ID No.

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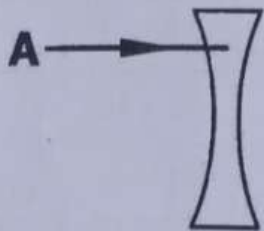
FOR EXAMINERS' USE ONLY		
QN. NO.	MARKS	EXR'S NO.
1 - 10		
11 - 20		
21 - 30		
31 - 40		
41 - 43		
44 - 46		
47 - 49		
50 - 52		
53 - 55		
TOTAL		

SECTION A: 40 MARKS

Questions **1** to **40** carry one mark each.

1. Name the part of a flowering plant which grows into a side branch.
.....
2. State any **one** factor to be considered when selecting a site for keeping honey bees.
.....
3. Give any **one** use of proteins in the human body.
.....
4. Give any **one** way in which evaporation is important in the environment.
.....
5. Why is it advisable to chew food properly before swallowing it?
.....
6. State any **one** way in which Athlete's foot can be controlled among school children.
.....

The diagram below shows a light ray **A** onto a lens. Use it to answer questions **7** and **8**.



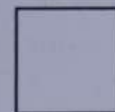
7. Name the type of lens in the diagram above.
.....
8. Complete the path of ray **A** as it passes through the lens.

9. Give any **one** role a School Health Club can play in the control of malaria in a community.

.....

10. Name the part of a human tooth that contains blood capillaries and nerve endings.

.....



11. State the reason why the volume of water in a container rises when a stone is lowered into it.

.....

12. How does cigarette smoking affect the human lungs?

.....

.....

13. State any **one** element of Primary Health Care (PHC) which is important in the prevention of COVID-19.

.....

14. Why are tomatoes grouped as annual crops?

.....

.....

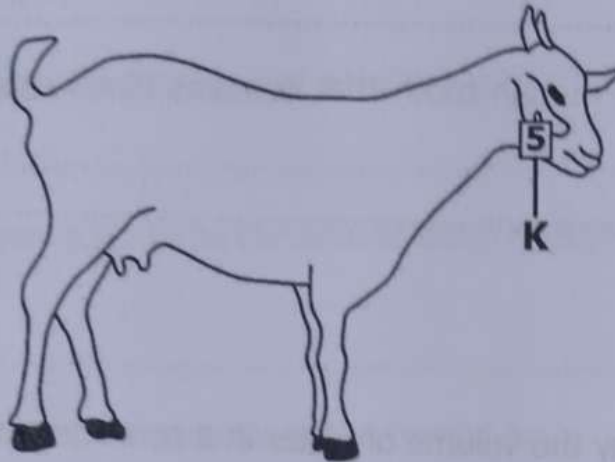
15. State any **one** way of conserving fossil fuels.

.....

16. Name the vector which spreads river blindness to people.

.....

The diagram below shows one of the management practices in goat rearing. Use it to answer questions **17** and **18**.



17. Name the management practice shown with letter **K**.
.....
18. How is the above practice important to a goat farmer?
.....
.....
19. Which type of weather favours proper harvesting of beans?
.....
20. Give any **one** way in which you can reduce chances of accidents when crossing a road.
.....
-
21. Name any **one** human body organ protected by the rib cage.
.....
22. Give any **one** condition that may make an electric bulb fail to produce light in a complete circuit.
.....

23. State any **one** way in which an adolescent can control bad odour that results from body changes.

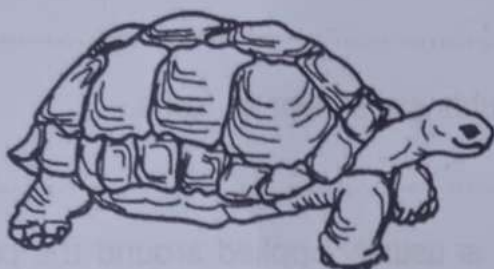
.....

.....

24. Mention any **one** way in which you can prevent some clothes from staining others during washing.

.....

Use the diagram of the animal below to answer questions **25** and **26**.



25. Identify the class of vertebrates to which the above animal belongs.

.....

26. How does the above animal protect itself?

.....

27. State any **one** way in which the spread of tuberculosis can be controlled.

.....

28. Mention any **one** way in which food may get contaminated after it has been prepared.

.....

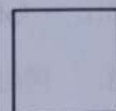
29. Why does sound travel fastest in solids compared to other states of matter?

.....

.....

30. Apart from proper insulation, state any **one** other way of protecting children against electric shock at home.

.....



31. Give any **one** way in which regular physical exercises help to maintain the muscular-skeletal system healthy.

.....

32. Give **one** way in which moulds are similar to ferns.

.....

33. State **one** reason why ash is usually applied around the pit hole of a latrine.

.....

.....

34. Name **one** excretory product removed by the lungs.

.....

35. Write down any **one** activity that members in a School Science Club can do to conserve the environment.

.....

36. State any **one** way in which echoes can be reduced in a large hall.

.....

37. State **one** way in which water is important during seed germination.

.....

38. How does tapeworm infestation lead to malnutrition in humans?

.....

39. Give any **one** way in which drug poisoning can be prevented among children at home.

.....

40. Give any **one** effect of cloud cover in an area.

.....

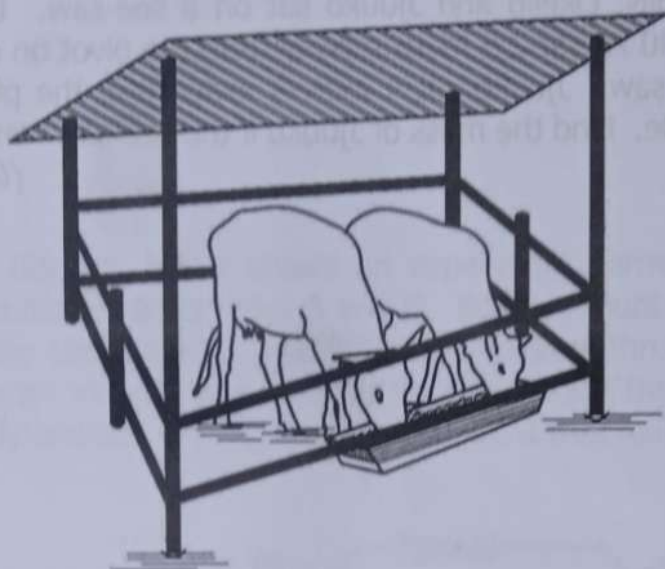


SECTION B: 60 MARKS

Questions **41** to **55** carry four marks each.

41. State the type of change that takes place when;
- (a) A knife rusts.
 - (b) Water freezes.
 - (c) A fruit ripens.
 - (d) Salt dissolves in water.
42. (a) Give the meaning of the following terms:
- (i) Sanitation.
.....
.....
 - (ii) Personal hygiene.
.....
.....
- (b) Identify any **two** activities children can do at school to promote good sanitation.
- (i)
 - (ii)

43. The diagram below shows a cattle grazing system. Study and use it to answer the questions that follow.



- (a) Name the cattle grazing system shown above.

.....

- (b) What type of natural manure can be got from the above system?

.....

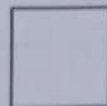
- (c) Apart from easy collection of manure, state any **two** advantages of the above cattle grazing system.

- (i)

.....

- (ii)

.....



44. (a) To which class of levers does a see-saw belong?

.....

(b) Two pupils, Okello and Jjuuko sat on a see-saw. Okello who weighs 40 kg sat 2.5 metres away from the pivot on one side of the see-saw. Jjuuko sat 4 meters away from the pivot on the other side. Find the mass of Jjuuko if the see-saw was balanced.
(03 marks)

45. (a) Name the largest vein in the human body.

.....

(b) Name the type of blood that is transported by the pulmonary artery.

.....

(c) State **one** characteristic of arteries that enables them to carry blood at high pressure.

.....

(d) Give **one** way in which the structure of capillaries enables them to exchange body materials.

.....

46. (a) Name the group of worms to which the following belong;

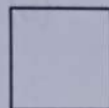
(i) Liver flukes.

(ii) Hookworms.

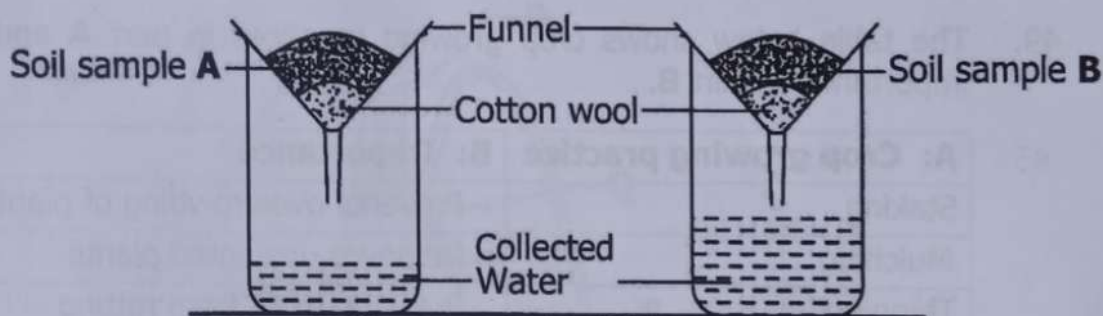
- (b) Apart from improving soil aeration, identify any other **two** ways in which earthworms are important in the soil.

(i)

(ii)



47. The diagram below shows an experiment carried out on two equal amounts of soil samples **A** and **B**. Equal amount of water was poured on the soil samples. The water that passed through the soil samples after ten minutes was collected in containers of the same size as shown. Study and use it to answer the questions that follow.



- (a) What property of soil was being studied in the experiment?

.....

- (b) State the function of the cotton wool in the experiment above.

.....

- (c) Given that the two soil samples were from sand and clay, which of the two samples was clay?

.....

- (d) Give any **one** reason for your answer in (c) above.

.....

.....

48. (a) State any **two** effects of chewing miraa (khat) to the human body.

(i)

(ii)

(b) Mention any **two** things a teenager can do to avoid the habit of chewing miraa (khat).

(i)

(ii)

49. The table below shows crop growing practices in part **A** and their importance in part **B**.

A: Crop growing practice	B: Importance
Staking	- Prevents overcrowding of plants
Mulching	- Removes unwanted plants
Thinning	- Prevents fruits from rotting
Weeding	- Conserves soil moisture

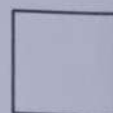
Match correctly the crop growing practices with their importance in the spaces provided below.

(i) Staking

(ii) Mulching

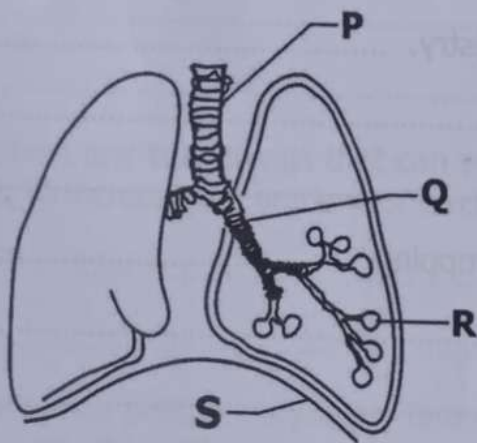
(iii) Thinning

(iv) Weeding



50. (a) Name the method of seed dispersal in;
- (i) Black jack.
- (ii) Coconut.
- (b) Give any **two** ways in which seed dispersal is important to plants.
- (i)
-
- (ii)
-

51. The diagram below is of a human respiratory system. Study the diagram and use it to answer the questions that follow.



- (a) Name the part marked **Q**.
-
- (b) Which substance is part **P** made of?
-
- (c) Give the function of the part marked **R**.
-
- (d) What happens to part **S** during the process of breathing in?
-

52. (a) Name the **two** states of matter through which heat travels by convection.

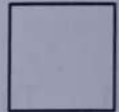
(i)

(ii)

(b) State any **two** ways in which heat transfer by convection is helpful in our daily life.

(i)

(ii)



53. (a) Give the meaning of the following terms:

(i) Agroforestry.

.....

.....

(ii) Mixed cropping.

.....

.....

(b) State any **two** characteristics of a tree suitable for agroforestry.

(i)

.....

(ii)

.....

54. You are provided with the following materials:

Salt, sugar, teaspoon, boiled water and clean containers. After washing your hands, write **four** other steps you will take to prepare salt sugar solution for treating dehydration at home.

(i)

.....

(ii)

.....

(iii)

.....

(iv)

.....

55. (a) Write down any **two** things that can be added to chicken feeds in order to increase the amount of calcium.

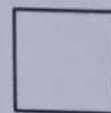
(i)

(ii)

(b) State any **two** practices a poultry farmer can do to layers to increase on egg production.

(i)

(ii)





UGANDA NATIONAL EXAMINATIONS BOARD
PRIMARY LEAVING EXAMINATIONS
2020

INTEGRATED SCIENCE

Time allowed: 2 hours 15 minutes

Index No.	Random No.						Personal No.		

Candidate's name :

Candidate's signature :

School Random number :

District No. :

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Read the following instructions carefully :

1. Do not write your **school** or **district name** anywhere on this paper.
2. This paper has two sections: **A** and **B**.
Section **A** has **40** questions and section **B** has **15** questions. The paper has **15 printed papers** altogether.
3. Answer **all** questions. **All** the answers for both sections **A** and **B** must be shown in the spaces provided.
4. **All** working must be done using a **blue** or **black** ball point pen or ink. Any work done in pencil will **NOT** be marked except drawings and diagrams.
5. Unnecessary **changes** in your work and handwriting that cannot easily be read may lead to loss of marks.
6. Do not fill anything in the table indicated: "**For examiners' use only**" and the boxes inside the question paper.

FOR EXAMINER'S USE ONLY		
Qn. No.	MARKS	EXR'S NO.
1 - 10		
11 - 20		
21 - 30		
31 - 40		
41 - 43		
44 - 46		
47 - 49		
50 - 52		
53 - 55		
TOTAL		

SECTION A: 40 MARKS

Questions **1** to **40** carry one mark each.

1. Name the type of cloud which is the highest in the sky.

.....

2. State any **one** way of controlling the spread of ring worm among children.

.....

3. Which type of soil erosion creates deep channels in the ground?

.....

4. Name the component of blood affected by the sickle cell disease.

.....

5. What is the function of the endosperm of a maize grain?

.....

The diagram below shows a change in state of matter. Use it to answer the questions **6** and **7**.



6. What is the state of matter at **Z**?

.....

7. Name the process by which **Z** changes back to liquid.

.....

8. Which part of a cow is affected by mastitis disease?

.....

9. State the principle on which a periscope works.

.....

10. State any **one** condition that favours the multiplication of bacteria.

.....



11. Give any **one** example of a non-renewable energy resource.

.....

12. State any **one** way in which the spread of germs through fingers can be controlled.

.....

13. Which method of heat transfer does not need any form of matter?

.....

14. State any **one** characteristic of a worker bee.

.....

The diagram below is of a musical instrument. Use it to answer questions **15** and **16**.



15. Name the group of musical instruments to which the instrument above belongs.

.....

16. How can the pitch of the above instrument be varied?

.....

.....

17. How are legumes different from other dicotyledonous plants?

.....

.....

18. State the method used to find the volume of an irregular object.

.....

19. Give any **one** way in which a school health club can promote health in their community.

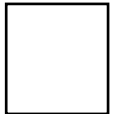
.....

.....

20. State any **one** advantage of having proper housing for goats.

.....

.....



21. How does the skin cool down the human body temperature on a hot day?

.....

.....

22. State any **one** danger of teenage pregnancy to adolescent girls.

.....

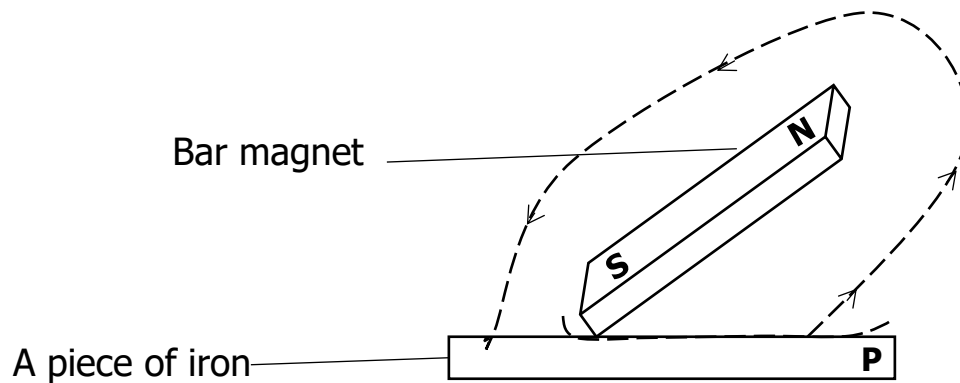
23. Give any **one** life skill that children need to develop in order to safe guard themselves against drug dependence.

.....

24. State any **one** way in which the spread of hookworms can be prevented.

.....

The diagram below shows a method of making a temporary magnet.
Study and use it to answer questions **25** and **26**.



25. Name the method shown above.

.....

26. What will be the pole at **P** after magnetization?

.....

27. How does chlorine make water safe for drinking?

.....

.....

28. Name the type of lens used in a magnifying glass.

.....

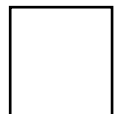
29. Apart from getting food, give any **one** other way in which rabbits benefit from plants.

.....

30. Why are triceps and biceps muscles referred to as voluntary muscles?

.....

.....



31. Apart from using medicine, state **one** other way of preventing the spread of coccidiosis in poultry.

.....

32. Give any **one** characteristic of a bird of prey.

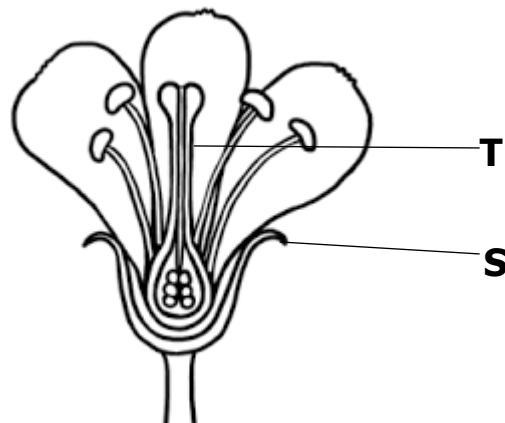
.....

33. How do cockroaches spread diseases to people?

.....

.....

The diagram below is of a flower. Use it to answer questions **34** and **35**.



34. Name the part labelled **T**.

.....

35. Give the function of the part labelled **S** to a flower before it opens up.

.....

36. Name the human respiratory disease whose signs are loss of weight, chronic cough and blood stained sputum.

.....

37. State the digestive disorder which can be prevented when people chew food properly and eat right amount.

.....

38. Give **one** reason why a person who has fainted should be placed in a cool and open space.

.....

39. Give **one** harmful effect of polythene bags to soil.

.....

.....

40. State the method which can be used to obtain salt that has dissolved in water.

.....



SECTION B: 60 MARKS

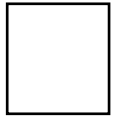
Questions **41** to **55** carry four marks each.

41. (a) What form of energy is stored in wood fuel?
.....
- (b) Name any **one** form of energy produced when wood burns.
.....
- (c) State **two** practices carried out at home to conserve wood fuel.
- (i)
- (ii)
42. (a) Apart from the vertebral column, give any **two** other parts of the Human skeleton used for protection.
- (i)
- (ii)
- (b) Name the part protected by the vertebral column in the human body.
.....
- (c) Give any **one** health habit that helps to strengthen the human body.
.....
.....
43. (a) Apart from animals, mention two other agents of seed dispersal.
- (i)
- (ii)

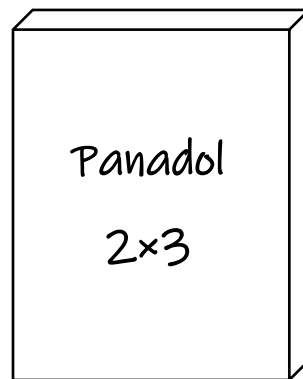
(b) State any **two** characteristics of seeds dispersed by animals.

(i)

(ii).....



44. A patient was given a packet of drugs labelled with information as shown in the diagram below. Use it to answer the questions that follow.



(a) What name is given to such information written on a packet of drugs?

.....

(b) Give any **one** reason why such information is important to patients.

.....

(c) How many times a day is the patient supposed to take the drug?

.....

(d) The patient took the drug in the morning, for how many hours should the patient wait before taking the drug again?

.....

45. (a) State any **two** effects of heat gain on matter.

(i)

(ii)

(b) Give any **two** ways in which a blacksmith uses heat

(i)

(ii)

46. The table below shows human body organs, the system to which they belong and the organ function. Study and complete it correctly.

ORGAN	SYSTEM	ORGAN FUNCTION
Heart	Circulatory	
.....	Excretory	Formation of urine.
Pancreas		Produces pancreatic juice
Epididymis	Reproduce	



47. (a) Name the group of simple machines to which each of the following tools belongs,

(i) a ladder

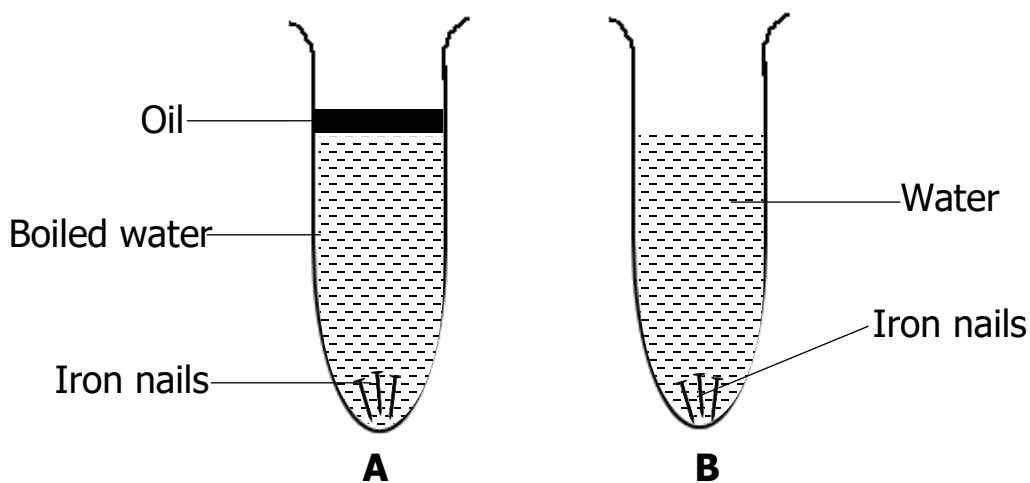
(ii) a panga

(b) Give any **two** ways in which simple machines make work easier.

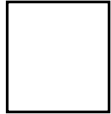
(i)

(ii).....

48. The diagram below shows an experiment about rusting. Use the diagram to answer the questions that follow.

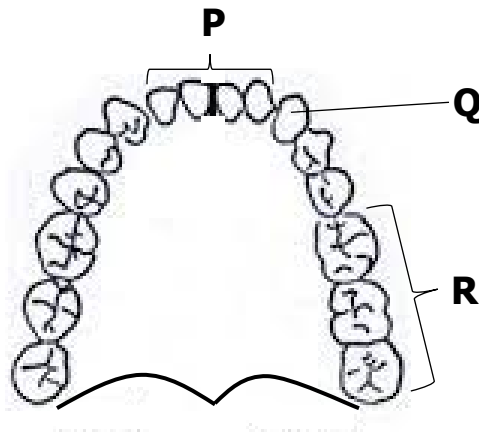


- (a) In which of the containers will the iron rust after some days?
.....
- (b) Why would the iron nails in the container you have identified rust?
.....
.....
- (c) Give **one** reason why oil was poured in container **A**.
.....
.....
- (d) What would happen to the nails if they were painted before being put in the containers?
.....
.....

49. (a) Name the type of immunity acquired through;
- (i) recovery from illness
 - (ii) immunization
- (b) Give any **two** roles of a family in promoting immunization in their area.
- (i)
.....
 - (ii)
.....
- 

50. (a) State **one** way in which cattle farming contributes to the development of leather industry.
-
- (b) Apart from leather industry, give any **one** other example of industry that uses cattle products.
-
- (c) State any **two** ways in which farmers can improve on the quality of cattle products.
- (i)
 - (ii)

51. The diagram below shows the arrangement of permanent teeth in the lower jaw of a human being. Study the diagram and use it to answer the questions that follow.



- (a) Name the type of teeth marked **Q**.

.....

- (b) State the function of the types of teeth labelled;

P:

Q:

- (c) Identify any **one** difference between the structure of the types of teeth labelled **Q** and **R**.

.....

.....

52. (a) Give any **two** reasons why farmers first plant some seeds in a nursery bed.

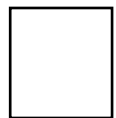
(i)

(ii)

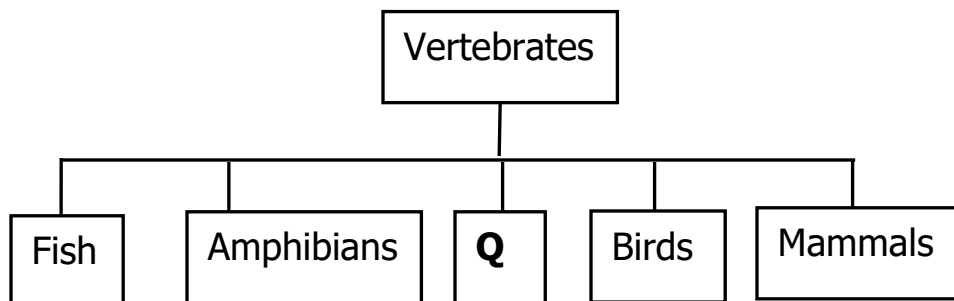
- (b) State any **two** advantages of planting crops in rows.

(i)

(ii)



53. The table below shows classes of vertebrates. Study and use it to answer the questions that follow.



- (a) Name the class of vertebrates represented by **Q**.

.....

- (b) Apart from having backbones, give **one** characteristic common to;

- (i) all fish, amphibians and vertebrates in class **Q**.

.....

- (ii) birds and mammals.

.....

- (c) State the difference in reproduction between vertebrates in class **Q** and the amphibians.

.....

.....

54. (a) Name the disease which results from the deficiency of vitamin C in human body.

.....

- (b) State any **two** signs of vitamin C deficiency disease in humans.

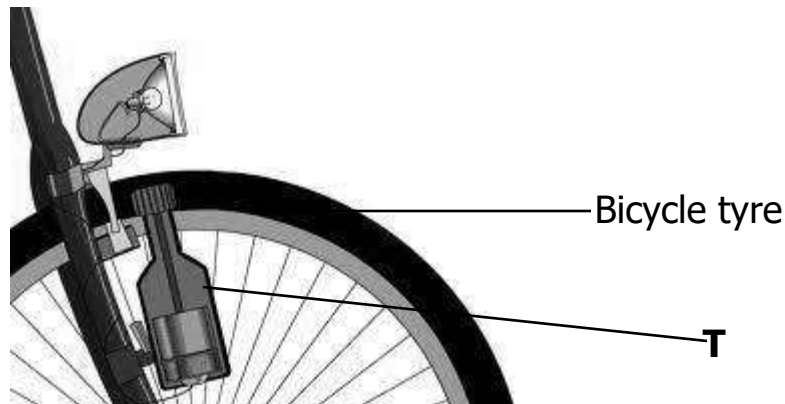
- (i)

- (ii)

- (c) Give any **one** example of food which gives us vitamin C.

.....

55. The diagram below shows an equipment on a bicycle that is used to produce electricity. Study and use it to answer the questions that follow.



- (a) Name the equipment labelled **T**.

.....

- (b) Give the function of the bicycle tyre in producing electricity.

.....

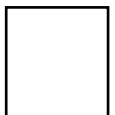
- (c) State the energy change that takes place in equipment labelled **T** when it is in use.

.....

- (d) State **one** way in which the amount of electricity produced by the equipment labelled **T** can be increased.

.....

.....





UGANDA NATIONAL EXAMINATIONS BOARD

PRIMARY LEAVING EXAMINATION

2019

INTEGRATED SCIENCE

Time Allowed: 2 hours 15 minutes

Index No.

Random No.						Personal No.		

Candidate's Name:

Candidate's Signature:

School Random No.

District ID:

Read the following instructions carefully:

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3. Answer **all** questions. **All** answers to both sections **A** and **B** must be written in the spaces provided.
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5. Unnecessary **changes** in your work and handwriting that cannot easily be read may lead to **loss of marks**.
6. Do **not** fill anything in the boxes indicated: **"For Examiners' use only"** and those inside the question paper.

FOR EXAMINERS' USE ONLY		
Qn. No.	MARKS	EXR'S NO.
1 - 10		
11 - 20		
21 - 30		
31 - 40		
41 - 43		
44 - 46		
47 - 49		
50 - 52		
53 - 55		
TOTAL		

SECTION A: 40 MARKS

Questions 1 to 40 carry one mark each.

1. Name the source of solar energy.
.....
2. Which vector spreads relapsing fever to people?
.....
3. State any **one** use of a razor blade in promoting personal hygiene.
.....
4. Give any **one** example of a biological change in plants.
.....
5. Which group of cold blooded vertebrates undergo internal fertilization?
.....

The diagram below is of a human skull and a neck vertebra. Use it to answer questions **6** and **7**.



6. Name the movable joint at **K**.
.....
7. Which delicate organ is protected by **S**?
.....

8. State the type of cattle kept to provide labour on a farm.

.....

9. What type of objects do not form shadows?

.....

10. Why does water drain faster in sand soil than clay soil?

.....

.....



11. How is the eardrum useful in the process of hearing?

.....

12. Write down any **one** activity done to seedlings during hardening off.

.....

13. State any **one** characteristic of incisor teeth which makes them suitable for their function.

.....

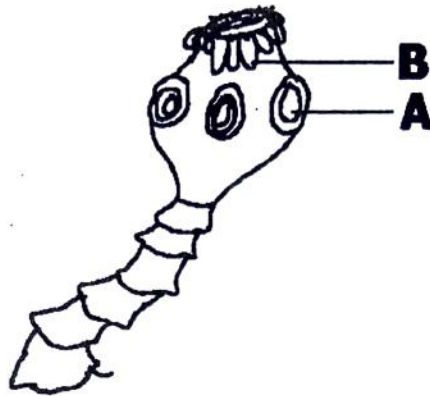
14. How can the breeding of houseflies be controlled in our compounds?

.....

15. Which deficiency disease is caused by lack of Vitamin B₁ in a diet?

.....

The diagram below shows a head of a tapeworm. Study it and answer questions **16** and **17**.



16. Name the part marked **A**.

.....

17. How is part **B** useful to the tapeworm?

.....

18. State any **one** characteristic of essential drugs.

.....

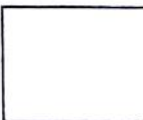
.....

19. Give any **one** way in which you can re-use plastic objects at home.

.....

20. How do carnivorous animals get energy from plants?

.....



21. State **one** characteristic of non-renewable resources.

.....

22. State any **one** difference in function between an artery and a vein.

.....

23. Give any **one** way in which having regular physical exercises is important to the human body.

.....

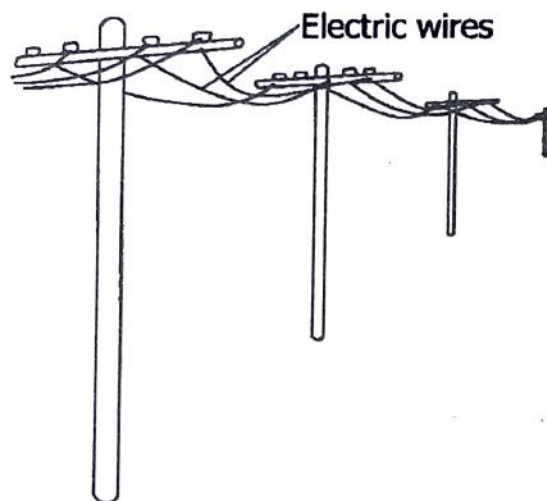
.....

24. How is a coconut seed adapted to water dispersal?

.....

.....

The diagram below shows electric wires under a certain weather condition. Study it and answer questions **25** and **26**.



25. In which kind of weather condition do the electric wires appear as shown above?

.....

26. Why do the electric wires appear as shown above?

.....

27. Why is alcohol used in a maximum and minimum thermometer?

.....

.....

28. How does mouth-to-mouth breathing help a victim of near drowning?

.....

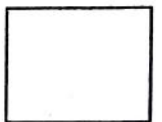
.....

29. State the importance of the hairs in the nose during breathing.

.....

30. Which immunisable disease affects the liver?

.....



31. Apart from Light Sussex, name any **one** other exotic breed of chicken kept for meat.

.....

32. Which rabbit disease causes wounds in their ears?

.....

33. Give any **one** way in which the electric current produced by a dynamo can be increased.

.....

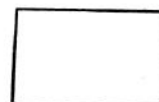
34. How does an object less dense than water behave in water?

.....

35. State any **one** condition which is favourable for bacteria to breed.

.....

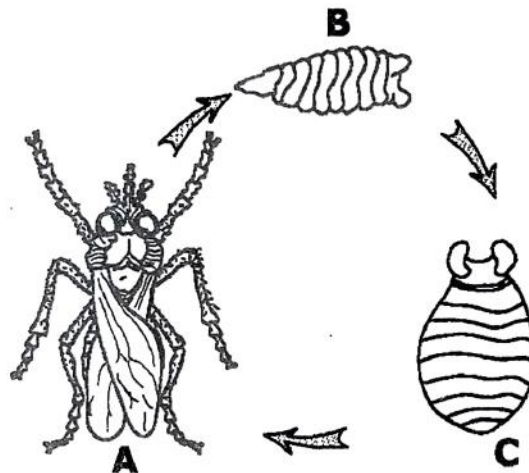
36. Name any **one** root tuber crop which is propagated by use of seeds.
.....
37. Which group of invertebrates has segmented bodies and jointed legs?
.....
38. How does the PIASCY programme help school children to avoid being infected with HIV/AIDS?
.....
.....
39. Give any **one** reason why farmers sort seeds before planting them.
.....
40. Apart from the skin, give any **one** other human body organ that excretes excess water.
.....



SECTION B: 60 MARKS

Questions 41 to 55 carry four marks each.

41. (a) Name the component of air used for;
- (i) respiration
 - (ii) putting out fire
- (b) State any **two** dangers of wind to people.
- (i)
 - (ii)
42. The diagram below shows the life cycle of a vector. Study it and answer the questions that follow.



- (a) Name the vector whose life cycle is shown above.
.....
- (b) Name the stage marked **B**.
.....
- (c) What does stage **A** feed on?
.....

- (d) How does the vector whose life cycle is shown above spread disease?

.....

43. (a) Write down any **two** causes of nose bleeding.

(i)

(ii)

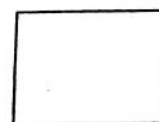
- (b) Why is a person whose nose is bleeding advised to;

(i) lean back his/her head?

.....

(ii) pinch the soft part of his/her nose?

.....



44. In the table below, part **A** shows terms used to describe sound and part **B** has the meanings of the terms.

Part A	Part B
Volume	Number of vibrations per second
Pitch	Reflected sound
Frequency	Loudness or softness of sound
Echo	Highness or lowness of sound

Use the meanings in part **B** of the table to match the terms below.

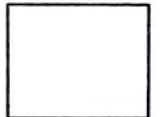
(a) Volume

(b) Pitch

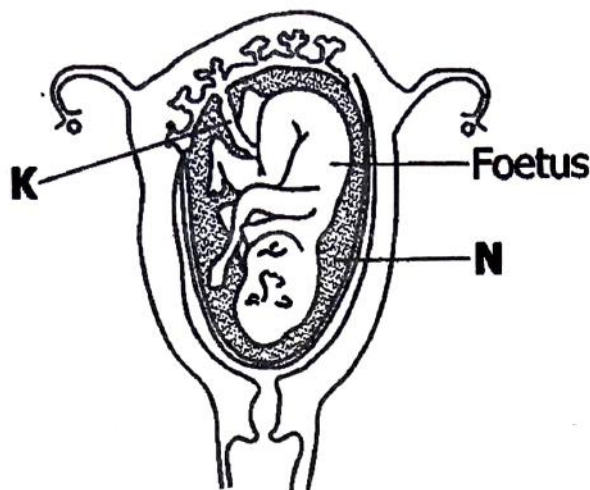
(c) Frequency

(d) Echo

45. (a) Write down any **two** reasons why toads are classified as amphibians.
- (i)
- (ii)
- (b) In which **two** ways are fish different from amphibians?
- (i)
- (ii)
46. (a) Write down any **two** pieces of information which are collected in a health survey.
- (i)
- (ii)
- (b) Give **two** reasons why carrying out a health survey in an area is important.
- (i)
- (ii)



47. The diagram below is of a developing foetus in a mother's womb. Study it and answer the questions that follow.



- (a) Name the substance marked **N**.

.....

- (b) Give **two** uses of the part marked **K** to the foetus.

(i)

(ii)

- (c) On the diagram above, label the placenta.

48. (a) Give **one** example of an exotic breed of goats kept in Uganda.

.....

- (b) Name **one** example of fibre got from goats.

.....

- (c) State any **two** reasons why some farmers would prefer rearing goats to cattle.

(i)

.....

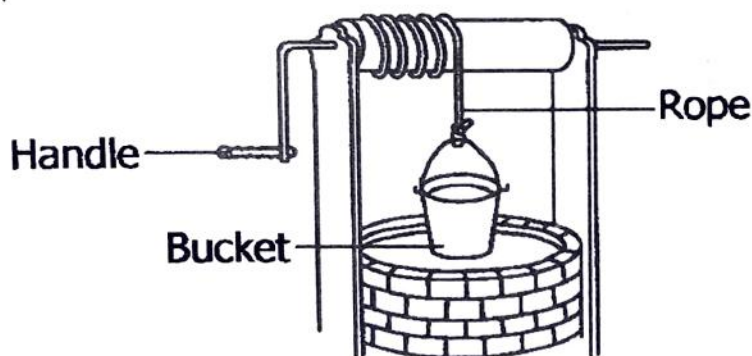
(ii)

.....

49. (a) State any **two** ways in which weeds can lower the yield of crops.
- (i)
- (ii)
- (b) State any **one** way in which the growth of weeds can be controlled in a garden.
-
- (c) Apart from controlling weeds, give any **one** other way farmers can care for their crops.
-



50. The diagram below shows a simple machine. Study it and answer the questions that follow.



- (a) To which type of simple machines does the machine above belong?
-
- (b) State any **one** use of the simple machine above.
-

- (c) Name any **one** force that makes work difficult when using the above machine.

.....

- (d) Give **one** other example of a machine that belongs to the above type of simple machine.

.....

51. (a) Which digestive juice is produced in the mouth?

.....

- (b) Write down any **two** things that happen to food in the mouth.

(i)

(ii)

- (c) Which digestive disorder is prevented by including roughage in our diet?

.....

52. (a) State any **two** factors that should be considered when starting a tree growing project.

(i)

(ii)

- (b) Give any **two** benefits of a woodlot to a community.

(i)

(ii)

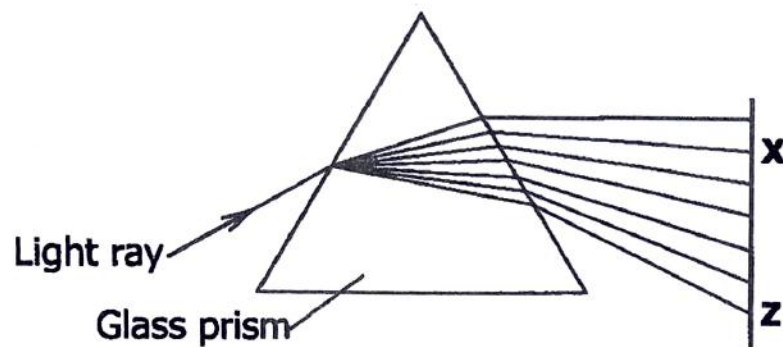


53. (a) What substance in plants helps them make their own food?

 (b) How is the feeding in ferns different from that of mushrooms?

 (c) Apart from ferns, give any **two** other plants which are similar to mushrooms in their way of reproduction.
 (i)
 (ii)

54. The diagram below shows how light is split into seven colours using a glass prism. Study it and answer the questions that follow.



- (a) Name the colours **X** and **Z**.
 (i) **X**
 (ii) **Z**
 (b) Why do the light rays bend at different angles as they leave the glass prism?

- (c) Why do we usually see the above band of seven colours in the sky when it is raining lightly?

.....

.....

55. (a) State any **two** signs of poor sanitation in a school.

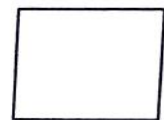
(i)

(ii)

- (b) Give any **two** diseases that can break out due to poor sanitation in a school.

(i)

(ii)



SCIENCE PLE 2018

CANDIDATE'S INFORMATION

Index number :

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Name :

Signature :

School name :

District name :

SECTION A: 40 MARKS

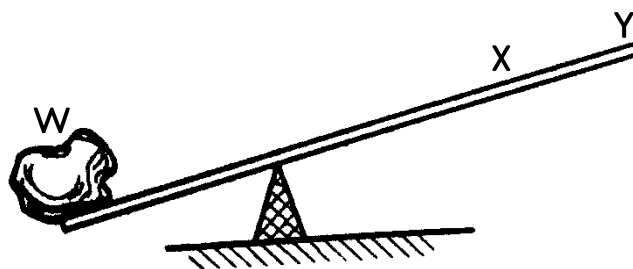
1. State any one sign of a compound fracture.

2. What happens when a new queen is born in a hive?

3. Which human body organ balances the amount of water needed in the body?

4. Give any one disorder of the human digestive system.

The diagram below is of a crow bar. Study and use it to answer questions 5 and 6.



5. Which of the two positions X and Y requires less effort to overcome the load W?

6. Indicate on the diagram with an arrow E, the direction of the effort used to overcome the load W.
7. Why do houseflies lay their eggs in decaying matter?

8. Why is a fuse made of a thin wire?

9. Where are seeds of coniferous plants found?

10. How is the function of anthers in flowers similar to that of testes in humans?

11. How do farmers control the breeding of poor quality animals on their farms?

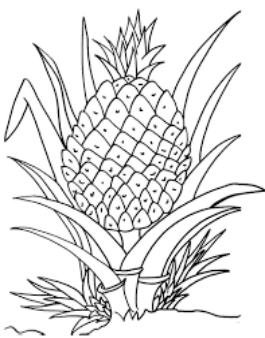
12. By what process is alcohol made from fruit juice?

13. Give any one way in which school children can promote immunisation in their community.

14. What First Aid would you give to a child who has taken paraffin?

15. State the importance of observing good posture to the human body.

The diagram below shows a fruit crop. Use it to answer questions 16 and 17.



16. How is the above crop propagated?

17. Give one other crop which is propagated in the same way as the crop shown above.

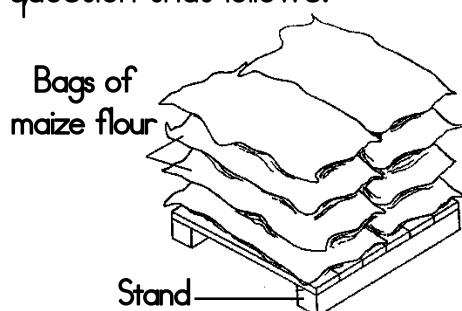
18. Give any one way in which human beings can prevent tapeworm infection.

19. How does smoking control the effect of bacteria in food preservation?

20. How are arteries able to withstand pressure of blood pumped from the heart?

21. State any one way in which people use sand soil.

22. Give any one disadvantage of overcrowding chicken in their house.
-
23. Which part of a maize grain absorbs and supplies food to the embryo?
-
24. What is the importance of the black lining inside a solar drier?
-
25. Mention any one factor considered when sorting clothes for washing.
-
26. Apart from removing dirt, why should the udder and teats be washed with warm water before milking?
-
27. How does prolonged breastfeeding help in child spacing?
-
28. What type of change is moulting in insects?
-
29. How is a lunar eclipse formed?
-
30. What determines the sinking or floating of an object in water?
-
31. How does covering of food during cooking help to conserve wood fuel?
-
32. State any one characteristic of images formed on the retina.
-
33. Why are eagles referred to as birds of prey?
-
34. The diagram below shows bags of maize flour in a store. Use it to answer the question that follows.



Why are the bags of maize flour put on such a raised stand?

35. How is water as a renewable resource replaced in the environment?

36. Give any one way in which proper disposal of wastes is important in our environment.

37. What would you do to a guitar string to make it produce a high pitched sound?

38. Why do the lungs expand during breathing in?

39. Give any one way in which the spread of Hepatitis B disease can be controlled.

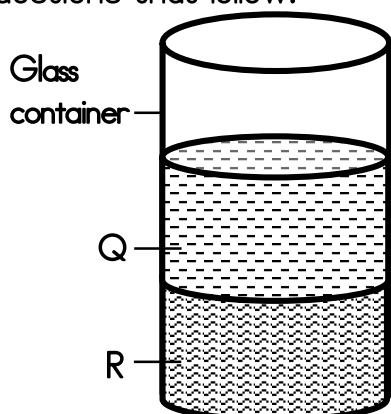
40. How are plants important in a food chain?

SECTION B: 60 MARKS

41. a. Name the outermost layer of the human skin.

- b. State any one waste product removed by the skin.

- c. Apart from removing wastes, mention two other functions of the human skin.
- i. _____
- ii. _____
42. The diagram below shows a glass container into which water and cooking oil were poured. The two liquids settled as shown. Study and use it to answer the questions that follow.



- a. Which of the two liquids is represented by:
Q _____
R _____
- b. State any one method that can be used to separate the two liquids.

- c. Why has liquid Q settled on top of liquid R?

43. In the table below, part A shows processes in human reproduction. Part B shows meaning of the processes in their wrong order.

A: Processes	B: Meaning
a. Ovulation	union of a sperm and an ovum.
b. Implantation	release of ovum from ovary.
c. Pregnancy	attachment of the embryo on the uterus wall.
d. Fertilization	a period from fertilization to birth.

Select from the table the correct meaning of the processes and write in the space provided below.

- a. Ovulation _____
- b. Implantation _____
- c. Pregnancy _____
- d. Fertilization _____

44. a. Write down any two causes of weathering of rocks.

- i. _____
- ii. _____

- b. Apart from weathering, name the other process by which soil is formed.

- c. How do living organisms in soil help to improve soil fertility?

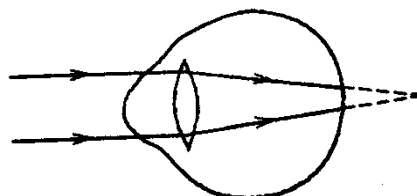
45. a. Give the importance of each of the following components of oral rehydration solution (ORS) to a patient.

- i. Sugar _____
- ii. Water _____

- b. Apart from using ORS, give two other ways of replacing lost fluids in a patient.

- i. _____
- ii. _____

46. The diagram below shows a defect of the human eye. Study and use it to answer the questions that follow.



a. Name the eye defect shown above.

b. State the two causes of the eye defect shown above.

i. _____

ii. _____

c. What type of lens is used to correct the above eye defect?

47. The table below shows weather elements, instruments and their uses. Study and complete it correctly.

Element	Instrument	Use
Rainfall	_____	Measures the amount of rainfall.
_____	Hygrometer	Measures the amount of water vapour in the atmosphere.
Wind	_____	Measures the speed of wind.
temperature	thermometer	_____ _____

48. a. State the two ways in which animals depend on other animals.

i. _____

ii. _____

b. Give any two ways in which plants depend on other plants.

i. _____

ii. _____

49. The following is a list of animals that belong to different classes of vertebrates. Use it to answer the questions that follow.

Monkey, Parrot, Crocodile, Tilapia

a. In which class of vertebrates does a crocodile belong?

b. Name the class of vertebrates which is not represented in the above list.

c. Identify any one animal from the above list which is cold blooded.

d. How is the reproduction of tilapia different from that of a parrot?

50. a. Give one reason why drugs should be kept;

i. Away from moisture and sunlight

ii. Out of reach of children

b. State any two ways in which people in a family can misuse drugs.

i. _____

ii. _____

51. a. What germ causes mastitis in cattle?

b. Give any two signs of mastitis.

i. _____

ii. _____

c. State any one way of controlling mastitis in cattle.

52. a. Give any two qualities of seeds that can germinate.

i. _____

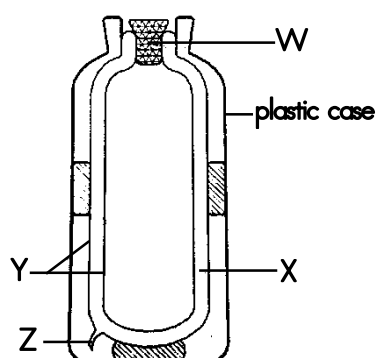
ii. _____

b. State any two farm practices which help in controlling insect pests in a garden.

i. _____

ii. _____

53. The diagram below is of a thermos flask. Study and use it to answer the questions that follow.



a. Name the parts labelled Y and Z.

Y _____

Z _____

b. Give any one material that can be used for making part W.

c. Why is part X able to prevent heat loss by conduction and convection?

54. a. State any two differences between the external parts of a hen and those of a cock.

i. _____

ii. _____

b. Write down any one product got from poultry birds.

c. Give any one way in which farmers can improve on the quality of their local breeds of chicken.

55. a. Give any two possible ways through which a baby can get HIV/AIDS from a mother.

i. _____

ii. _____

b. State any two ways in which an HIV positive pregnant woman can protect her unborn baby from getting HIV/AIDS.

i. _____

ii. _____

SCIENCE PLE 2017

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SECTION A: 40 MARKS

56. State the injury caused by steam to a human body.

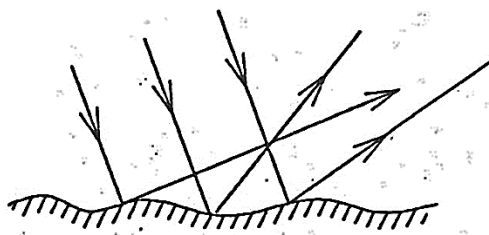
57. What type of electricity is lightening?

58. Give one way in which termites benefit from soil.

59. State one effect of gonorrhoea on the female reproductive system.

60. What is the importance of gap filling in crop growing?

The diagram below shows reflection of light. Study and use it to answer questions 6 and 7.



61. What type of reflection is shown above?

62. Why are the rays reflected as shown in the diagram?

63. State any one way in which alcoholism affects a family.
-
64. State any one reason why animals make sound.
-
65. Name the process through which water turns into ice?
-
66. In which one way does tooth paste promote oral health?
-
67. State any one physical change that takes place during the process of distillation of alcohol.
-
68. What is the importance of the rings of cartilage of the trachea in the respiratory system?
-
69. Why does a nail sink in water?
-
70. Give any one danger of wetland drainage to water animals.
-
71. State any one principle of Primary Health Care.
-
72. Why are laying boxes put in dark corners of a poultry house?
-
73. Apart from causing diseases, give one other way fungi are harmful to people.
-
74. How is a reflected sound helpful to sailors?
-
75. Give one way in which molluscs are different from other invertebrates.
-
76. Give any one mineral deficiency disease in humans.
-

77. Name the type of bees which develops from unfertilized eggs.

78. State any one way in which the spread of germs by houseflies can be controlled.

The diagram below shows a germinating seed. Study and use it to answer questions 24 and 25.



79. What type of germination is shown in the diagram above?

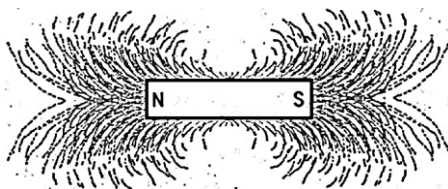
80. To which group of flowering plants does the germinating seed belong?

81. State any one way in which saliva is useful in the process of digestion.

82. Name the vaccine which is given to prevent three childhood immunisable diseases.

83. Give one reason why less effort is applied to move a load using first class levers.

84. The diagram below shows a bar magnet with iron filings around it. Study and use it to answer the question that follows.



Which property of magnets is shown in the diagram?

85. State any one sign of disease on plant leaves.

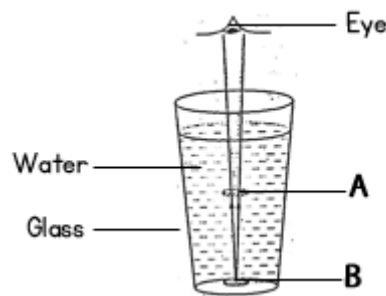
86. Name the organ which regulates the human body temperature.

87. State any one way in which animals can be used as energy resources.

88. Which method of harvesting trees allows new shoots to grow from the stump?

89. Give any one way of controlling the bad smell in an ordinary pit latrine.

The diagram below shows an object in a glass of water. Study and use it to answer the question that follows.



90. Why does the object seem to appear at point A yet it is at point B?

91. State any one sign of the snuffle disease in rabbits.

92. Which property of air makes the lungs expand when you breathe in?

93. Apart from curing diseases, give one other characteristic of essential drugs.

94. Why is bio gas regarded as an environmentally friendly source of energy?

95. State any one characteristic of seeds dispersed by wind.

SECTION B: 60 MARKS

96. a. Give any two signs which show that a baby is dehydrated.

i. _____

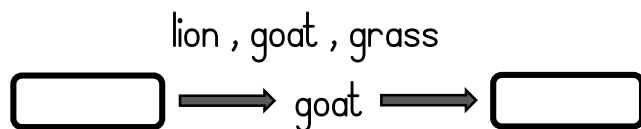
ii. _____

b. State any two pieces of advice you would give to a mother whose baby is dehydrated.

i. _____

ii. _____

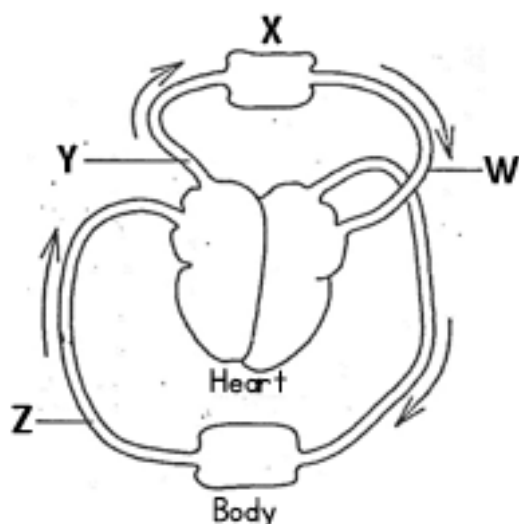
97. a. Use the living things listed below to complete the given food chain.



- b. Which one of the living things in the food chain above is a producer?

- c. What is the source of energy for the producer in the food chain above?

98. The diagram below shows the circulation of blood in the human body. Study and use it to answer the questions that follow.



- a. Name the organ marked with letter X.

- b. Which blood vessel is marked with letter W?

- c. State the similarity between blood vessel Y and blood vessel Z.

- d. Why does blood vessel W bring blood back to the heart?

99. a. Write down any two causes of soil erosion.

i.

ii.

- b. Which agent of soil erosion causes gullies?

- c. How does soil erosion lead to soil infertility?

100. The table below shows the information collected from five homes during a health survey. Study it to answer the questions that follow.

Homes	Sanitation items checked			
	Latrine	Rubbish pit	Bathing shelter	Drying rack
A	×	×	✓	×
B	✓	✓	✓	✓
C	×	✓	✓	✓
D	×	×	×	✓
E	×	✓	✓	×

- a. Which home had all the sanitation items?

- b. Identify the sanitation item missing in most of the homes surveyed.

- c. Name any one disease that is likely to attack homes A, C, D and E.

- d. Give any one other sanitation item which could be added in the health survey.

101. a. What causes fainting?

- b. State any two conditions that can lead to fainting.
- i. _____
- ii. _____
- c. Why are the legs of a person who has fainted raised higher than the head when giving first aid?

102. Use the list of crops given below to answer the questions that follow:

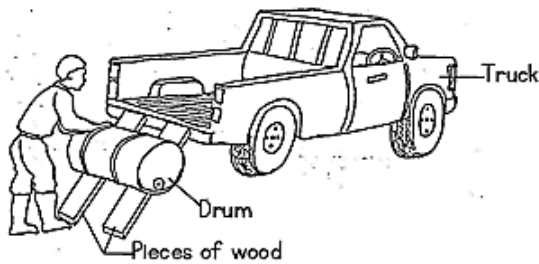
conifer, coffee, cassava, ground nut

- a. Which crop on the list is propagated by use of stem cutting?

- b. Identify any one crop on the list which is;
- An annual crop _____
 - A perennial crop _____

c. Give any one way in which a conifer is different from all other crops on the list.

103. The diagram below shows a man using a simple machine to load a drum onto a truck. Study and use it to answer the questions that follow.



a. What type of simple machine is the man using?

b. Give any two examples of the type of simple machine you have named in (a) above.

i. _____

ii. _____

c. How can the man improve on the simple machine above in order to make his work easier?

104. a. Which breed of cattle is obtained by mating a pure local breed and an exotic breed?

b. Give any two reasons why some farmers prefer keeping local breeds of cattle to exotic breeds.

i. _____

ii. _____

c. State any one characteristic of dairy cattle.

105. a. Name any two bones that form the joint at the knee of the human body.

i. _____

ii. _____

b. What name is given to the joint at the knee?

c. In which one way is friction reduced at a joint?

106. The table below shows some common infections, organisms that cause it and the body parts affected. Study and complete it correctly.

Common infectious	Organisms that cause it	Body parts affected
_____	Chlamydia	Eyes
Tuberculosis	Bacteria	_____
_____	Virus	Bones and muscles
Scabies	_____	Skin

107. a. Apart from rainfall, give one other element of weather necessary for plant growth.

- b. In which two ways is rainfall important to plant growth?

i. _____

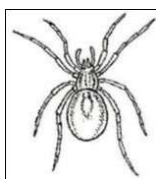
ii. _____

- c. State any one disadvantage of too much rainfall to plants.

108. The diagrams below show animals in the arthropods group. Study and use them to answer the questions that follow.



X



Y



Z

- a. Name the group of arthropods to which animals X and Y belong.

X _____

Y _____

- b. Give any one reason why animal Y does not belong to the same group as animal Z.

- c. How are animals X, Y and Z similar in the way they reproduce?

109. a. Apart from raising the temperature of substances, give any other two effects of heat on matter.

i. _____

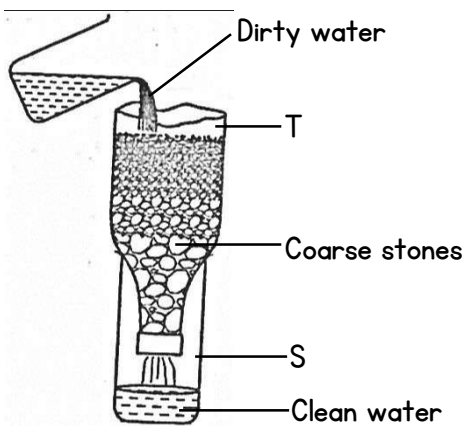
ii. _____

b. State any two ways in which heat transfer by conduction is important to people.

i. _____

ii. _____

110. The diagram below shows a simple method of making water clean. Study and use it to answer the questions that follow.



a. Name the method shown above.

b. Apart from the coarse stones, name any two other local materials which are placed in container T.

i. _____

ii. _____

c. Why should the clean water obtained in container S be boiled before drinking?

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SECTION A: 40 MARKS

1. Name the gas used in fire extinguishers.

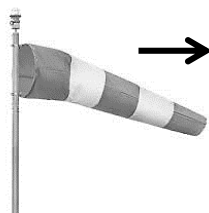
2. State the eye disease spread by houseflies.

3. Which part of a fish performs a similar function as the human ear?

4. Name the male reproductive cells in flowers.

5. Give any one way in which wood fuel can be conserved.

6. The diagram below is of a weather instrument. Use it to answer the question that follows.

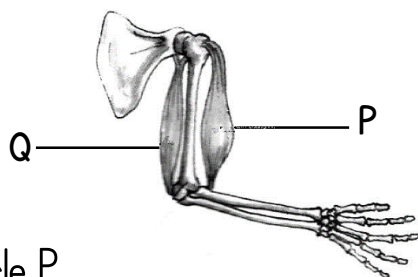


What is the use of this instrument at the weather station?

7. Which method of heat transfers enables a person to iron clothes using a flat iron?

8. Why should cold water be poured on the part of the body which has been scalded or burnt?
-
9. Apart from decomposition, in which other way is soil formed?
-
10. What is the importance of smoke during the harvesting of honey?
-
11. State the class of lever in which the effort is between the fulcrum and the load.
-
12. Give a reason why male anopheles mosquitoes do not transmit malaria parasites to humans.
-
13. How can foot and mouth disease be controlled in cattle?
-
14. Give one way in which farmers can improve on the supply of nitrogen in the soil without the use of artificial fertilisers.
-

The diagram below is of a human arm. Study and use it to answer questions 15 and 16.



15. Name muscle P.
-
16. What happens to muscle Q when the arm is raised.
-
17. Which intestinal worm enters the human body through bare feet?
-
18. How do plants make use of the gas given out by animals during respiration?
-

19. State any one use of plane mirrors to people.
-
20. Apart from helping in pollination, give one other way in which wind is useful to plants.
-
21. Which part of a maize grain has a similar function as the cotyledon of a bean seed?
-
22. State any one characteristic which shows that a bat is a mammal.
-
23. Which blood cells carry oxygen in the body?
-
24. Give any one reason why breast feeding is encouraged for at least two years?
-
25. Name any one scale in which temperature is measured.
-
26. The diagram below shows two bar magnets placed close to each other. Use it to answer the question that follows.



Use arrows to show what happens to the two bar magnets.

-
27. Name the human body system to which the lungs, kidneys and the skin belong.
-
28. Give any one way in which bacteria in pit latrines are useful.
-
29. Name one disease that affects the stomach because of alcoholism.
-
30. Apart from polio vaccine, name any other vaccine administered to children at 6 weeks of age.
-

31. In which part of the human alimentary canal is water absorbed?

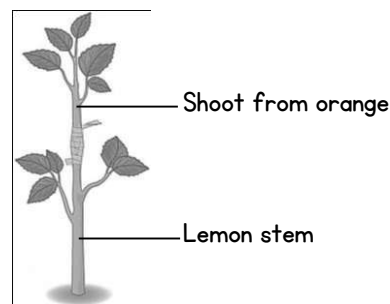
32. What is the danger of having too much wax in the human ear?

33. What form of energy is produced by dry cells?

34. Give any one good health habit that can improve the working of the lungs.

35. Apart from eating a balanced diet, give any one other way of controlling common diseases at home.

36. The diagram below shows a method of plant propagation. Use it to answer the question that follows.



Name the method of plant propagation shown in the diagram above.

37. Which organs are used by both fish and tadpoles for breathing?

38. Give any one secondary sex characteristic which is common to both adolescent boys and girls.

39. Name the fungi which normally grows on left over foods.

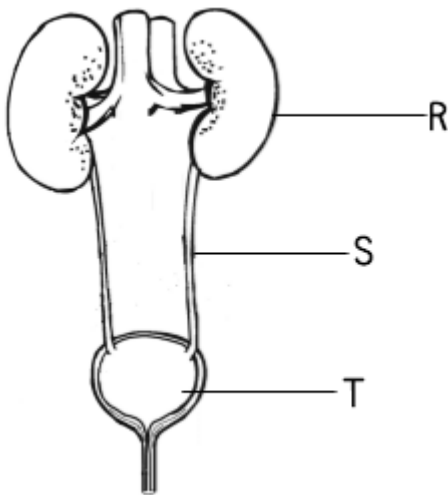
40. What happens to light rays when they meet a convex lens?

SECTION B: 60 MARKS

(Questions 41 to 55 carry 4 marks each)

41. a. State any two ways in which vectors spread diseases to humans.
- _____
 - _____
- b. Give any two ways of preventing vectors from spreading diseases to humans.
- _____
 - _____

42. The diagram below is of a human urinary system. Study and use it to answer the questions that follow.



- a. Name the parts marked S and T
- S _____
- T _____
- b. What is the function of part marked R.
- _____
- _____
- c. Give any one disease that affects the above system.
- _____

43. In the table below, part A shows activities carried out by farmers and part B shows the effects of the above activities.

Part A (activities)	Part B (effects)
Irrigation	preserves moisture.
Afforestation	leads to the death of crop pests.
Mulching	promotes convectional rainfall.
Crop rotation	allows growth of crops in all seasons.

Write the correct effect to the activity in the space provided below:

- Irrigation _____
- Afforestation _____
- Mulching _____
- Crop rotation _____

44. a. Give any two examples of non-renewable resources in the environment.
- _____
 - _____
- b. State any two ways of conserving non-renewable resources.
- _____
 - _____
45. a. To which group of plants do ferns and liverworts belong?

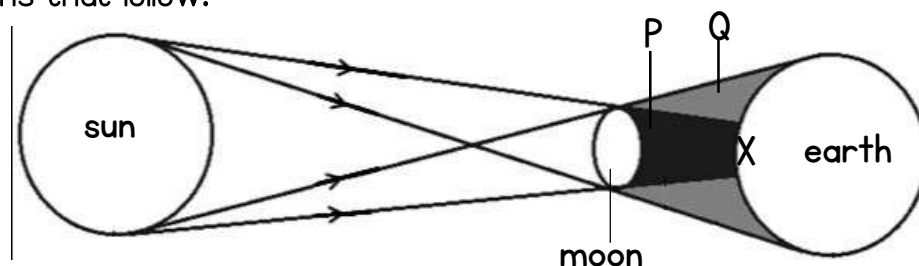
b. How is the reproduction of ferns similar to that of liverworts?

c. Give two other plants that reproduce in the same way as ferns and liverworts.

- _____
- _____

46. a. Name the change of state which takes place when;
- Vapour changes to liquid. _____
 - A solid changes directly to gas. _____
- b. Apart from causing change of state, give two other effects of heat on matter.
- _____
 - _____

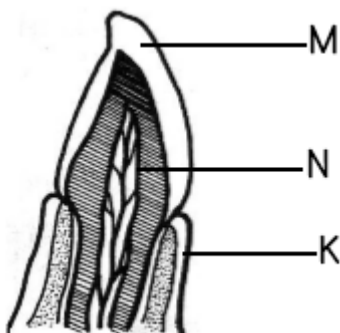
47. The diagram below shows a type of eclipse. Study and use it to answer the questions that follow.



- a. Name the type of eclipse shown in the diagram above.
- _____
- b. Name the shadows marked P and Q.
- P _____
- Q _____
- c. What happens to a person who would be in part X during the eclipse?
- _____

48. a. In which one way is each of the following elements of Primary Health Care (PHC) important to the community?
- Water and sanitation _____
 - Maternal and child health _____
- b. Give any two health care services which are provided through maternal and child health.
- i. _____
 - ii. _____
49. a. Name two systems of keeping poultry that enables a farmer to keep large number of birds.
- i. _____
 - ii. _____
- b. State any two viral diseases that can spread easily in the above systems of poultry keeping.
- i. _____
 - ii. _____
50. a. Give any one characteristic which is common to all invertebrates.
- _____
- b. To which group of invertebrates does of the following animals belong?
- Snail _____
 - Grasshopper _____
- c. State one way in which snails may be dangerous to humans.
- _____

51. The diagram below shows a section through a human tooth. Study and use it to answer the questions that follow.



- a. Name the parts marked M and K

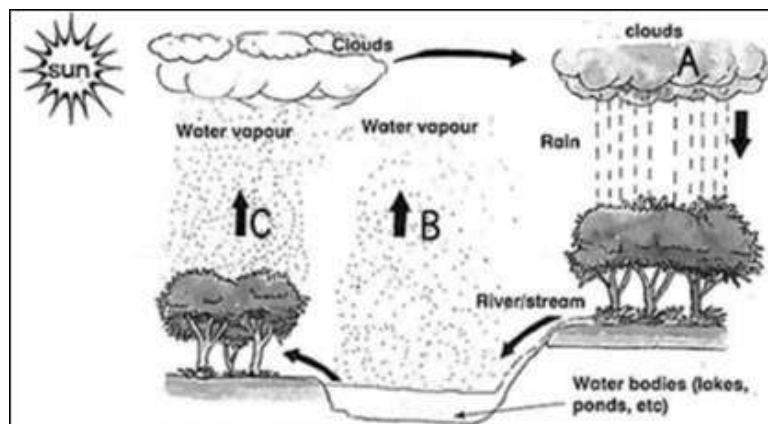
M _____

K _____

- b. Which mineral salt helps to form part M?

- c. Give the function of part labelled N.

52. a. State the method that can be used to separate;
- fruit seeds from juice. _____
 - Millet seeds mixed with its husks. _____
- b. Give the importance of the following steps in cleaning clothes at home.
- Sorting _____
 - Soaking _____
53. a. Write down any two examples of anti-social behaviours in our community.
- i. _____
 - ii. _____
- b. State any two ways of controlling anti-social behaviours in our community.
- i. _____
 - ii. _____
54. a. Apart from using solfa notation, mention any two other ways of storing sound.
- i. _____
 - ii. _____
- b. Name two instruments used to reproduce the sound stored by solfa notation.
- i. _____
 - ii. _____
55. The diagram below shows the water cycle. Study and use it to answer questions that follow.



- a. Name the type of clouds represented by letter A.
- _____
- b. State the process taking place at B and C.
- B _____
- C _____
- c. What is the importance of the sun in the above diagram?
- _____

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SECTION A: 40 MARKS

1. Which type of immunity do babies get from their mothers?

2. Name the blood vessel which takes blood from the heart to the lungs.

3. Name the energy resource obtained from animal dung and urine.

4. Mention any one factor that affects the rate of evaporation.

5. State the difference in units used to measure weight and mass.

The diagram below shows a simple machine. Use it to answer questions 6 and 7.



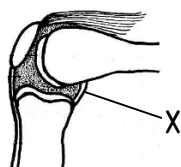
6. To which class of levers does the machine above belong?

7. Use an arrow with letter P to show the fulcrum on the diagram.
8. Give any one way in which litter is important in a poultry house.

9. Apart from the bright coloured petals, state one other thing that attracts insects to flowers.

10. State any one way in which drought affects food production in the community.
-
11. State the difference between the feeding of a tapeworm and a hookworm.
-
12. Why is water obtained by distillation method not good for drinking?
-
13. State the sense organ for feeding in the human body.
-
14. Which property of magnets enables a magnetic compass to work?
-
15. State any one way which the spread of scabies can be prevented.
-
16. Why do some plants shed their leaves during dry season?
-
17. Mention one permanent method of birth control in males.
-
18. Why does a fish die soon after it has been removed from water?
-
19. Give the use of a drenching gun to a cattle farmer.
-
20. Give one reason why pregnant women are advised to feed on food rich in proteins.
-

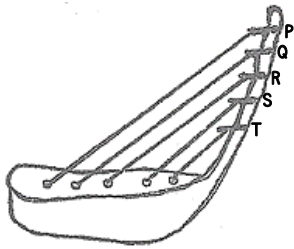
The diagram below shows parts of a joint. Study and use it to answer questions 21 and 22.



21. Name the part marked with letter X.
-
22. Using letter W, label the part that reduces friction in the joint, apart from the synovial fluid.

23. State the method of controlling soil erosion where farmers plough land across a slope.
-
24. Which element of Primary Health Care helps to prevent tooth decay?
-
25. State the role of enzymes during food digestion.
-
26. Apart from cameras and microscopes, name any one other optical instrument that uses a lens.
-
27. Give one way in which mosses are similar to mushrooms in the way they reproduce.
-
28. Why is tepid sponging an important First Aid to a person with fever?
-
29. Why are sun birds able to suck nectar from the bottom of flowers?
-
30. Give an example of mollusk which is a vector to humans.
-
31. Name the human body system which eliminates wastes.
-
32. Give any one way in which a VIP latrine is different from an ordinary pit latrine.
-
33. Name the root crop which is attacked by the mosaic disease.
-
34. Give any one reason why sheep farmers carry out docking.
-
35. Apart from a flat iron, give any one other appliance used at home that changes electric energy to heat energy.
-

36. The diagram below shows a well-tuned musical instrument. The strings P, Q, R, S and T are of uniform thickness. Use it to answer the question that follows.



Why does string T produce sound of the highest pitch?

37. State the any one way in which smoking affects the human respiratory system.
-

38. Give the importance of plants in a food chain.
-

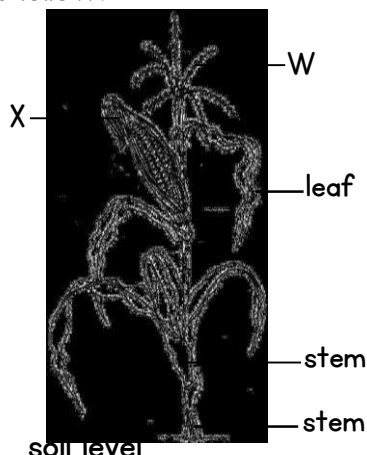
39. Name the fungal caused sexually transmitted disease, which is spread through poor hygiene.
-

40. State one reason why farmers stake tomato plants.
-

SECTION B: 60 MARKS

Each question carries 4 marks

41. a. Give two examples of natural fertilizers.
- i. _____
- ii. _____
- b. State any two ways in which natural fertilizers are better than artificial fertilizers.
- i. _____
- ii. _____
42. The diagram shows a flowering plant. Study and use it to answer the questions that follow.



- a. To which group of flowering plants does the above plant belong?
-
- b. What type of root system does the plant have?
-

c. State the importance of the type of root marked Y to the plant.

d. In which one way is part W different from X in their reproductive function?

43. a. Give any two conditions in which women are vulnerable.

- i. _____
- ii. _____

b. State any two ways in which family members can care for vulnerable women.

- i. _____
- ii. _____

44. a. Name the respiratory organs which are protected by the rib cage.

b. What happens to the rib cage during the following stage processes?

- Breathing in _____
- Breathing out _____

c. Give the use of oxygen we breathe in.

45. The table below shows diseases, the vectors that spread them and the disease causing germs. Study and complete it correctly.

Disease	Vector	Germ
Cholera	Houseflies	_____
Yellow fever	_____	Virus
_____	Infected cats	Virus
River blindness	Black flies	_____

46. a. Give the function of each of the following parts of a modern beehive:

- The brood chamber _____
- The honey chamber _____

b. A part from being used as food, give two other ways in which honey is useful to people.

i. _____

ii. _____

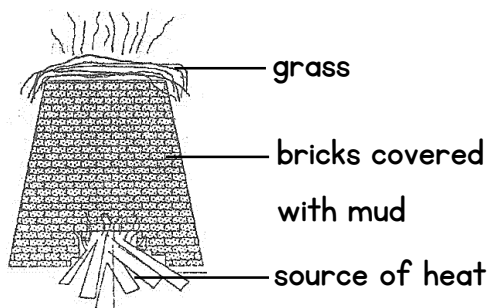
47. a. What kind of food is made by green plants during photosynthesis?

b. State any one raw material used by green plants for photosynthesis.

c. How is chlorophyll important during the process of photosynthesis?

d. Give one reason why photosynthesis does take place at night.

48. The diagram below shows firing bricks in a skin. Study it and use it to answer the questions that follow.



a. Name the source of heat energy used in firing the bricks

b. Give any one reason why the bricks are covered with mud before firing starts.

c. What is the importance of the grass put on top of the kiln during firing?

d. State any one disadvantage of using the above method of firing bricks to the environment.

49. The table below shows parts of a human eye in A and those of a camera in B.

A	B
Iris	Shutter
Pupil	Film
Eye lid	Diaphragm
Retina	Aperture

For each of the parts of the human eye, write the part of the lens camera from B which performs a similar function.

- a. Iris _____
- b. Pupil _____
- c. Eye lid _____
- d. Retina _____

50. a. Which communicable disease shows signs and symptoms similar to that of tuberculosis?

b. Name the one cattle product through which tuberculosis can spread to human beings.

c. When is the vaccine against tuberculosis administered in babies?

d. Apart from immunization, give one other way in which the spread of tuberculosis can be controlled in humans.

51. a. Give any two ways in which alcohol is used in hospitals.

- i. _____
- ii. _____

b. State any two effects of alcoholism to an individual.

- i. _____
- ii. _____

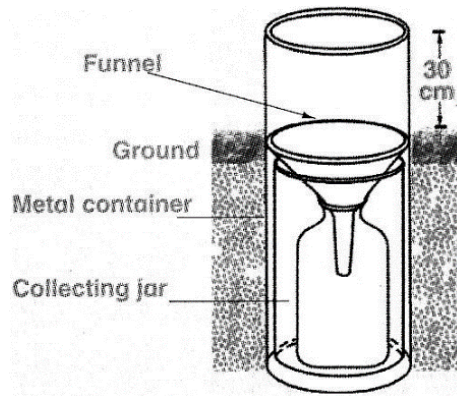
52. a. Apart from the respiratory and body wastes, name any other two materials carried in blood.

- i. _____
- ii. _____

b. Name the component of blood which transport oxygen in the human body.

c. Which disease causing germ attacks the white blood cells in humans?

53. The diagram below shows a weather instrument. Use it to answer the questions that follow.



- a. Name the weather instruction shown in the diagram above.

- b. What is the use of the instrument above at weather station?

- c. Give reason why people are advised to:

- Put the instrument in an open ground when using it.

- Raise the instrument 30cm above the ground when using it.

54. a. To which class of vertebrates do toads belong?

- b. Give one way in which the breathing of a toad is different from that of a tadpole.

- c. State any two ways in which the jelly round the egg of toads is important.

55. a. Apart from ticks, name any one other cattle pest.

- b. State any two diseases spread by ticks to cattle.

- c. Give any one way in which the tick borne diseases can be controlled.

SCIENCE PLE 2014

CANDIDATE'S INFORMATION

Index number :

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Name :

Signature :

School name :

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SECTION A: 40 MARKS

1. Which type of type soil has rough and large particles?

2. In which state of matter does heat travel slowest?

3. To which class of levers does a wheelbarrow belong?

4. Apart from labour, give one other requirement for starting a livestock farm.

5. What kind of accident requires the use of splints in giving the First Aid?

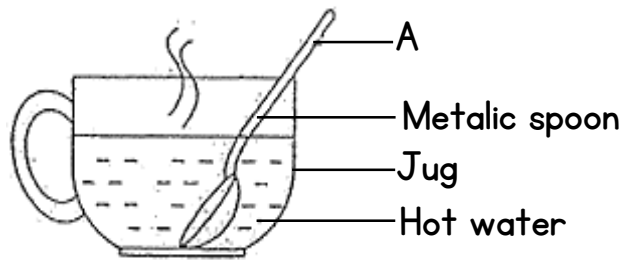
6. Give any one exotic breed of sheep kept for wool production.

7. Which sexually transmitted disease can cause blindness in babies?

8. Name the human body organ that uses light energy for it to function.

9. State one way in which echo can be reduced in a hall.

10. The diagram below shows a piece of metal (spoon) dipped in a jug containing hot water. Use it to answer the question that follows.



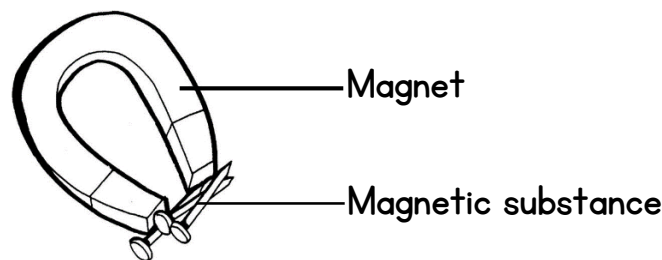
How does part A become hot yet it is not in hot water?

11. Give any one disadvantage of harvesting honey using fire.
-
12. How is the BCG vaccine different from the polio vaccine in the way they are given to children?
-
13. Give one reason why washing of hands after visiting a latrine is a good practice.
-
14. Which process in plants has a similar function as sweating in humans?
-
15. Apart from taking wrong dose, give one other way in which drugs are misused.
-
16. Which condition in babies is shown by sunken soft on the head?
-
17. Give one way in which plants differ from animals for photosynthesis.
-
18. How is sun-drying helpful in the preservation of food crops?
-
19. Why is water grouped under renewable resources?
-
20. Which mineral helps in the formation of red blood cells in the human body?
-

21. A part from melting, state any one other physical change that takes place in the presence of heat.

22. Name any other one disease of rabbits that affects poultry birds.

23. Name the method of making a magnet shown in the diagram below.



24. In which method of making water clean, why do you first allow the impurities to settle at the bottom of the container?

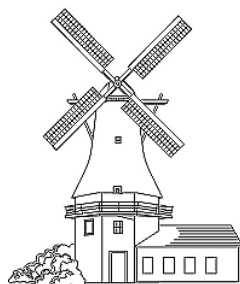
25. Why does a coin thrown in a basin full of water appear to be nearer to the surface?

26. How is the reproduction in whales different from that of fish?

27. Why is it important for government to collect information on human health?

28. State any one piece of advice you would give to a person who suffer from indigestion.

29. The diagram below shows how electricity is produced. Use it to answer the question that follows.



What source of energy is used to produce the electricity?

30. Why does paraffin settle on top when mixed with water?
-
31. Give any one way in which bacteria are useful to crop farmers.
-
32. Mention the group of birds to which a crested crane belongs.
-
33. Which disease is spread to human by rats?
-
34. How is the function of the pulmonary vein different from that of other veins?
-
35. Which property of air enables you to drink soda using a straw?
-
36. Why do plane mirrors produce regular reflection of light?
-
37. What name is given to a meal containing all the food values in their right amounts?
-
38. State one way in which married couples can avoid HIV/AIDS among themselves.
-
39. Give one way in which the use of solar energy is good to our environment.
-
40. Apart from the number of body parts, state one other difference between arachnids and insects.
-

SECTION B: 60 MARKS

41. a. Name two poisonous substances found in tobacco.

i. _____

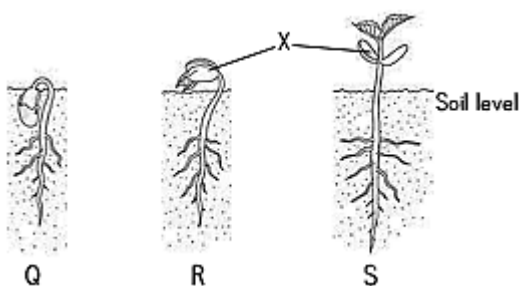
ii. _____

b. State any two effects of cigarette smoking by pregnant women to their unborn babies.

i. _____

ii. _____

42. In the diagram below, Q, R and S show the stages of germination of a seed. Study and use it to answer the questions that follow.



a. Name the part marked X.

b. How is the part marked X useful to the seedling at stage S?

c. Why is the part marked X useless at stage S?

d. What type of germination is shown by the seed?

43. The table below shows joints and their position in the human body. Study and complete it correctly.

Joints	Position in the body
Pivot joint	_____
_____	Shoulder
_____	Knee
Suture joint	_____

44. a. State any two agents of soil erosion.

i. _____

ii. _____

b. Give two ways in which soil erosion can be controlled in hilly areas.

i. _____

ii. _____

45. a. What type of change takes place when an iron tool rusts?

b. Give two conditions necessary for rusting to take place on an iron tool.

i. _____

ii. _____

c. State any one effect of rusting on iron tools.

46. a. Apart from egg eating, name any other vice in poultry.

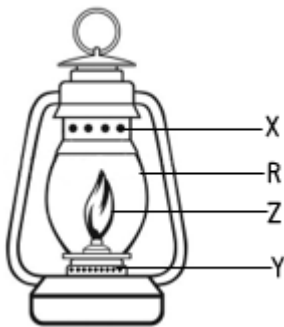
b. Give any one cause of egg eating in poultry.

c. State any two ways of preventing the egg eating vice in poultry.

i. _____

ii. _____

47. The diagram below shows a kerosene lamp. Study and use it to answer the questions that follow.



a. Name the part marked with letter Z.

b. Why is the part marked R made of glass?

c. Give the importance of the parts marked X and Y when the lamp is in use.

X _____

Y _____

48. a. Name the group of crops that are harvested year after year.

b. Give any two examples of crops that belong to the group you have named in (a) above.

i. _____

ii. _____

c. State one way in which the above crops are harvested.

49. The table below shows a group of animals A with their reparatory organs in B.

A	B
Tilapia	Moist skin
Mosquito	Lungs
Frog	Gills
Dog	Spiracles

Write against each animal below, its respiratory organ from B.

a. Tilapia _____

b. Mosquito _____

c. Frog _____

d. Dog _____

50. a. In the human reproductive system, state where each of the following process take place:

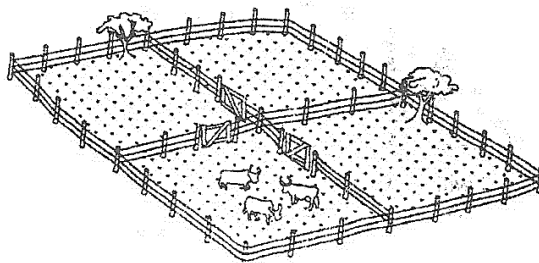
- Implantation _____

- Production of sperms _____

- Production of female hormones _____

b. Give any one use of the placenta to the foetus during pregnancy.

51. The diagram below shows a system of grazing cattle. Study it and use it to answer the questions follow.



a. Name the system of grazing cattle shown in the diagram above.

b. Besides restricting the animals, gives two other advantages of this system of grazing cattle.

i. _____

ii. _____

c. State any one disadvantage of the system of grazing cattle shown in the diagram.

52. a. Give two ways in which Primary Health Care (PHC) is an important programme in the community.

i. _____

ii. _____

b. State any two roles of a school health committee.

i. _____

ii. _____

53. a. What name is given to the practice of growing crops and trees together on the same piece of land?

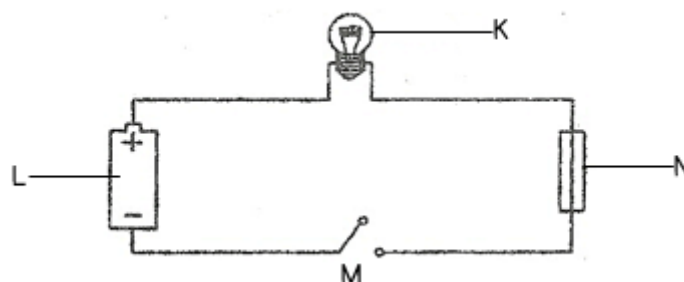
b. State any two ways in which crops benefit from trees when they are grown together.

i. _____

ii. _____

c. Give any one proper method of harvesting trees in the practice names in (a) above.

54. Study the diagram of an electric circuit below and use it to answer the questions that follow.



a. Name the parts marked K and N.

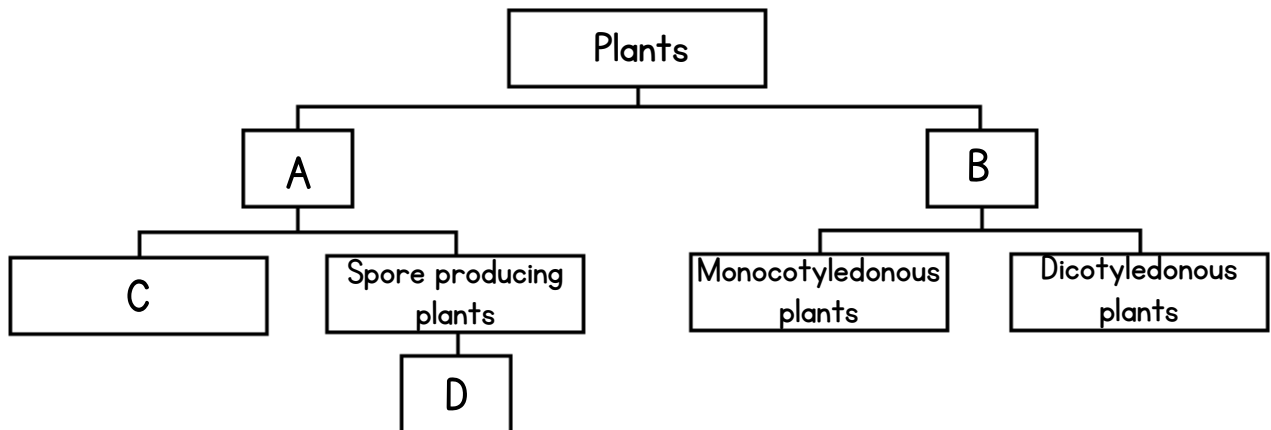
K _____

N _____

b. State the energy change that takes place at L when M is closed.

c. Give any one form of energy produced at K when M is closed.

55. The table below shows classification of plants. Study it carefully and use it to answer the questions that follow.



a. Name the group of plants shown by the following letters.

A _____

B _____

C _____

b. Give any one plant that belongs to the box labeled D.

SCIENCE PLE 2013

CANDIDATE'S INFORMATION

Index number :

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Name :

Signature :

School name :

District name :

SECTION A: 40 MARKS

1. Give any one example of a swimming bird.

2. Name the animal product used in leather industry.

3. State any one property of matter.

4. Name the part of the human eye that works like the film in the Camera.

5. In which way is the message in the sign post below helpful to school children?

SAY NO TO SEX

6. Give any one way in which single fixed pulley is used a school.

7. Name the body organ which helps a person to know whether there is enough sugar in tea or not.

8. Give any one advantage of pruning trees.

9. Which method of heat transfer helps farmers to dry their harvested crops?

10. How does a fish benefit from its slippery body?

11. In which way does use of biogas conserve the environment?

The diagram below shows garden tool. Use it to answer questions 12 and 13.



12. Name the garden tool above.

13. How is the tool above important to farmers?

14. Which deficiency disease is caused by lack of vitamin D in children's diet?

15. Why is the water obtained through filtration not necessary safe for drinking?

16. What plant system does the plumule make when it grows?

17. How does leaching reduce soil fertility?

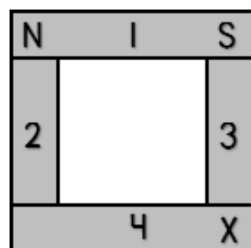
18. Why is it advisable to leave the pit of a VIP latrine open?

19. Name the cattle disease whose sign is a swollen and painful udder.

20. In which part of the female reproductive system does fertilization take place?

21. Why is the left side of the heart able to pump blood to most body parts?

22. State any one way in which trees are able to reduce air pollution.
-
23. In the space provided below, draw a lens used to correct short sightedness.
-
24. How is a safety pin in a First-Aid kit important?
-
25. In which type of change are properties of the substance formed permanently different from those of the original one?
-
26. How is the function of stomata in a leaf similar to that of gills in a fish?
-
27. Which component of soil is used by plants to make food?
-
28. Which structures enable the trachea to remain open all the time?
-
29. Four bar magnets marked 1, 2, 3 and 4 were placed next to each other as shown below. They attracted each other. Name the pole of magnet 4 marked with letter x.
-



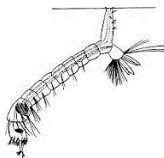
30. Apart from brushing, how else can dental caries be prevented?
-
31. State any one way in which friction in moving parts of a machine can be reduced.
-
32. Give any one way in which yeast is useful to people.
-

33. A part from making food, mention one other use of leaves to plants.

34. State any one characteristic of the images formed by plane mirrors.

35. Why do the people installing electricity wear rubber gloves?

The diagram below is of mosquito larva. Use it to answer questions 36 and 37.



36. Name the mosquito which develops from such a larva.

37. What disease does the mosquito you have named above transmit to human beings?

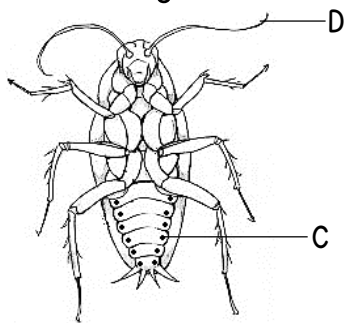
38. How is the propagation of Irish potatoes different from that of sweet potatoes?

39. A part from running windmills, give one other way wind is used as an energy resource.

40. How do the silvery walls of a vacuum flask reduce heat loss?

SECTION B: 60 MARKS

41. The diagram below shows an insect. Use it to answer the questions that follow.



a. Name the parts marked C and D.

C

D

b. State the function of part marked D.

c. Why would an insect die when the whole of its abdomen is dipped in oil?

42. a. A part from the lack of conditions necessary for germination, give any two other factors that can make a seed fail to germinate.

i. _____

ii. _____

b. Write down any two ways farmers can control pests in the garden without using chemicals.

i. _____

ii. _____

43. a. Give any two advantages of modern beehive.

i. _____

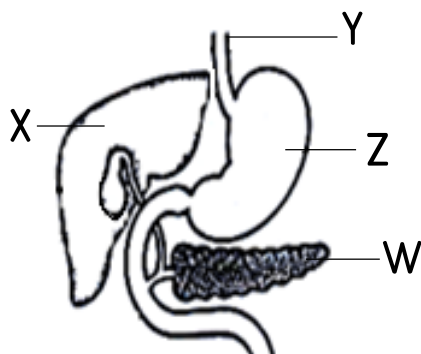
ii. _____

b. Write down two things a bee-keeper should consider before setting up beehives.

i. _____

ii. _____

44. The diagram below is part of the human digestive system. Use it to answer the questions that follow.



a. Name the parts marked Y and W.

Y _____

W _____

b. State the function of the part marked Z.

c. Give any one disease that affects the part marked X.

45. Match the disease in list A with the signs and symptoms in list B.

List A

Polio

Measles

Tetanus

Diphtheria

List B

swollen neck

stiffness of muscles

paralyzed tissue

skin rash

- a. Polio _____
- b. Measles _____
- c. Tetanus _____
- d. Diphtheria _____

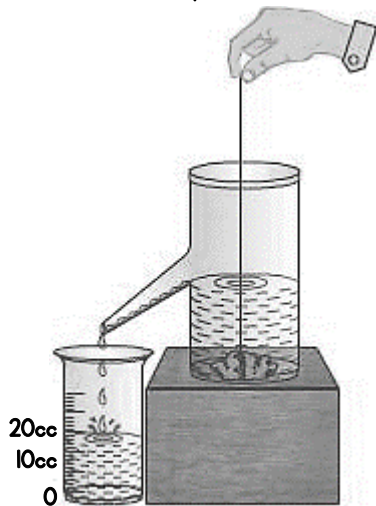
46. a. Give any two ways in which drug dependence can affect an individual.

- i. _____
- ii. _____

b. State two ways in which an individual can avoid drug dependence.

- i. _____
- ii. _____

47. An irregular object was lowered into an overflow can A containing water. The water it displaced was collected in container B as shown below.



a. What is the volume of the irregular object?

b. Calculate the density of the irregular object if its mass is 60g.

48. The table below lists of animals. Use it to answer the questions that follow.

P	Q	R
Toad	Chameleon	Cow
Newt	Snake	Monkey
Frog	Lizard	Rat

a. Which characteristic is common to all the animals in the table?

b. How is the reproduction in the animals in list P similar to those in list Q?

c. Which list has warm blooded animals?

d. Why would you place a tortoise in list Q?

49. a. Name any one organism that makes fresh meat go bad.

b. State any three ways of preventing fresh meat from going bad.

i. _____

ii. _____

iii. _____

50. a. Name the human body organ where each of the following takes place.

– Filtration of blood

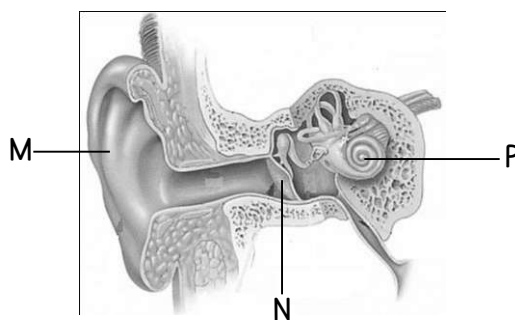
– Blood gets oxygen while carbon dioxide is removed

b. Give the use of the following components of blood in the body.

– White blood cells

– Blood platelets

51. Study the diagram of the human ear below and use it to answer the questions that follow.



a. Name the parts marked M and N.

M _____

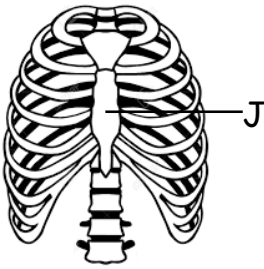
N _____

b. Give the function of the marked P

c. Use letter B to show the part of the ear responsible for balancing the body.

52. a. Name any two water-borne diseases.
- _____
 - _____
- b. In which two ways can water-borne diseases be prevented in the community?
- _____
 - _____

53. The diagram below shows part of the human skeleton. Use it to answer the questions that follow.



- a. Name the part marked J.

- b. State any two delicate body organs protected by the part of the skeleton shown.

- _____
- _____

- c. How is the part of the human skeleton shown above used during breathing out?

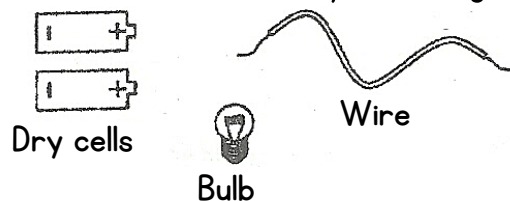
54. a. Write down any two activities that can be done by an individual to promote personal hygiene.

- _____
- _____

- b. State two ways community members can promote sanitation in their area.

- _____
- _____

55. Arrange the parts an electric circuit below by drawing to show a complete circuit.



SCIENCE PLE 2012

CANDIDATE'S INFORMATION

Index number :

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Name :

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SECTION A: 40 MARKS

1. Give any one example of root tuber.

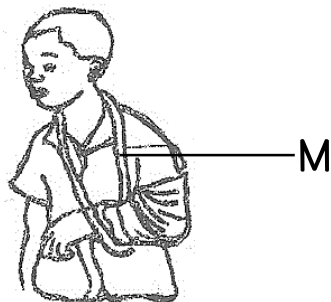
2. State one disadvantage of keeping finger nails long.

3. What is the importance of including calcium in chicken feed?

4. What is the use of a wedge as a simple machine?

5. Name the type of lens used to correct long sightedness.

The diagram below shows a person with an injury. Use it to answer question 6.

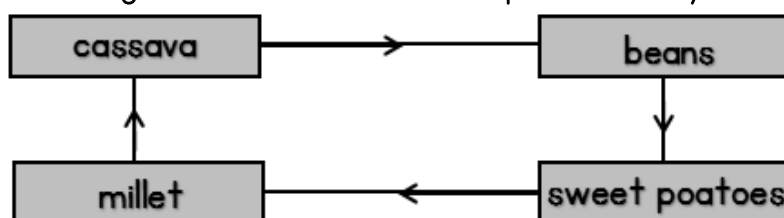


6. How useful is the structure marked M to the injured person?

7. Apart from mosses, give one other example of a spore bearing non flowering plant.

8. Name any one mammal that lives in water.
-
9. Give a reason why people keep cats in their homes.
-
10. Why does a farmer cut off the leaves of banana sucker before planting it?
-
11. Apart from pasteurizing, give any one way milk can be prevented from going bad.
-
12. Why is it important for husband and his wife to be treated together if they have got gonorrhea?
-
13. Why do farmers carry out carry out de-beaking in poultry farms?
-
14. State any one habit which helps to promote oral health.
-
15. State one use of soil to earthworms.
-

The diagram below shows a crop rotation cycle.



16. What is the importance of beans in this cycle to the crops?
-
17. Which communicable disease in human affects both the skin and the skeletal system?
-
18. Apart from carbohydrates, name one other class of food that gives the body energy.
-
19. In which way is the sun important in the formation of rain?
-

20. State any one method used to dirty water clean.

21. How does a sweater help to keep a person warm during cold day?

The diagram below is of a head of a bird. Use it to answer question 22.



22. What does a bird with this type of beak eat?

23. Give any one effect of poor disposal of human wastes in the environment.

24. Apart from the egg stage, at what other stage of development is a housefly not active?

25. Why does a clay-made charcoal stove use less charcoal than a metallic one?

26. Apart from production of egg (ova), mention one other use of the ovaries in the human body.

27. State one way in which a child gets natural immunity.

28. State the injury that results from tearing of muscles.

29. Which gas is used by food manufacturers for preserving drinks like sodas?

The diagram below is of a common animal. Use it to answer question 30.



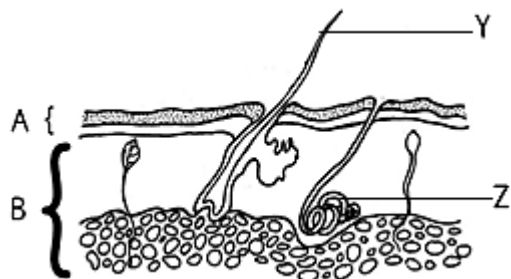
30. Which disease does the above animal spread to humans?

31. Why are the legs of a person who has fainted raised higher than the head as a way of giving first aid?
-
32. What name is given to a place where seedling are grown before transplanting?
-
33. Which part of the human ear equalizes pressure in and outside the ear?
-
34. Name one other disease caused by fungi in humans apart from ring worm.
-
35. What class of food is digested in the stomach?
-
36. Give any one advantage of castrating farm animals.
-
37. In the space below, draw a simple diagram of a fibrous root system.
-
38. How are plastics and glass materials dangerous to soil?
-
39. Apart from liquids, name one other state of matter through which heat is transferred by convection.
-
40. In which human body organ does gaseous exchange take place?
-

SECTION B: 60 MARKS

41. a. Give any two examples of a physical change.
- i. _____
- ii. _____
- b. State two characteristics of a chemical change.
- i. _____
- ii. _____

42. The diagram below is of a human skin. Use it to answer the questions that follow.



a. Name the layers A and B.

A _____
B _____

b. Give the function of the parts labeled Y and Z.

Y _____
Z _____

43. a. Name the two types of pollination in flowering plants.

i. _____
ii. _____

b. Write any two agents of pollination.

i. _____
ii. _____

44. a. Give any two disadvantages of keeping local breeds of poultry.

i. _____
ii. _____

b. Apart from getting money, give any two other ways a farmer benefits from keeping poultry.

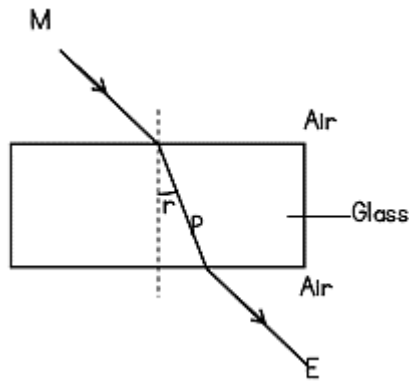
i. _____
ii. _____

45. Match each animal in A with the correct group to which it belongs in B.

A	B
Chameleon	mammals
Toad	birds
Bat	reptiles
Ostrich	amphibians

- a. Chameleon _____
b. Toad _____
c. Bat _____
d. Ostrich _____

46. The diagram below shows how light passes through a glass block. Study and use it to answer question 46.



- a. Name the rays M and P.

M _____

P _____

- b. Name the angle marked r.

- c. Give one reason why ray P bends as shown in the diagram.

47. a. Give one reason why dehydration is dangerous to humans.

- b. In which way is oral Rehydration Solution (ORS) a good treatment for diarrhea?

- c. Apart from taking ORS, how else can one prevent dehydration?

- d. How can a person preparing ORS prevent it from contamination?

48. The table below shows source of food, the food nutrient in it and the related deficiency disease. Complete it correctly.

Source of food	Food nutrient	Deficiency disease
Fruits	vitamin C	_____
Beans	_____	kwashiorkor
_____	vitamin A	night blindness
Iodized salt	iodine	_____

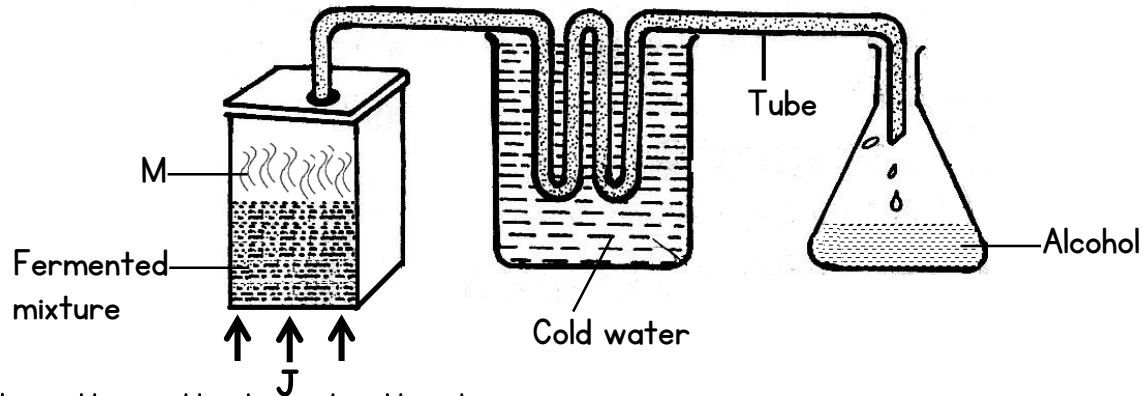
49. a. Which type of blood vessels returns blood to the heart?

- b. What is the function of valves in the blood vessels during blood circulation?

c. What type of blood is carried by most blood vessels with valves?

d. Give any one waste material carried by blood.

50. The diagram below shows one of the methods of preparing alcohol. Study and use it to answer question 50.



a. Name the method used in the diagram.

b. What do arrows labeled J represents?

c. Why is the tube passed through cold water?

d. What process forms M?

51. a. A part from ante-natal care, give two other ways of caring for pregnant women.

i. _____

ii. _____

b. In which one way should pregnant protect unborn babies from malaria?

c. Why are pregnant women advised not smoke?

52. a. Name the force that brings downs all objects thrown up.

b. (i). When a piece of paper and rubber are dropped from the same height, which one reaches the ground first?

(ii). Give a reason for your answer in (b) (i) above.

c. Why do some objects float on water?

53. a. State any two reasons why farmers thin their crop.

i. _____

ii. _____

b. Apart from thinning, give any two other ways farmers care for their crops.

i. _____

ii. _____

54. a. Name the type of joint found in each of the following parts of the human body.

– Shoulder _____

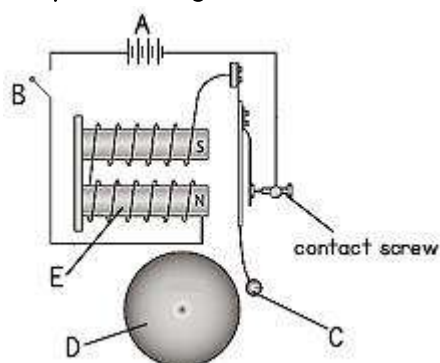
– Wrist _____

b. What advice would you give to a person to keep the skeletal system in the health working condition? (give two)

i. _____

ii. _____

55. Study the diagram of an electric bell below and use it to answer question 55.



a. Name the part marked A and C.

A _____

C _____

b. What will happen to E when B is closed?

c. How useful is part D on the electric bell?

SCIENCE PLE 2011

CANDIDATE'S INFORMATION

Index number :

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Name :

Signature :

School name :

District name :

SECTION A: 40 MARKS

1. Name the disease which is spread by a female anopheles mosquito.

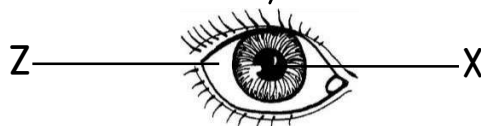
2. Which form of energy changes ice to water?

3. Give reason why drugs should be kept away from children.

4. Name the class of simple machines to which an axe belongs.

5. Write down any one example of a plant fiber.

The diagram below is of human eye. Use it to answer questions 6 and 7



6. Name the part marked with letter X.

7. Give the use of the part marked with letter Z.

8. State one type of accident whose First Aid is to put the affected part of the body in cold water.

9. Why should children be given food which is rich in proteins?

10. Give one way in which plants depend on soil.

11. Give one way in which the umbilical cord is important to the development to the developing fetus in the Uterus.

12. Name any one characteristic of seeds which are dispersed by wind.

The diagram below shows a fracture. Use it to answer question 13.



13. What type of fracture is shown in the diagram above?

14. State the compass direction in which a freely suspended magnet will rest.

15. What is the use of grills in a mushroom?

16. Give one way in which P7 pupils can promote immunization in their families.

17. Apart from pregnant mothers, young and disabled people, give any one other group of people who need special care in a family.

18. In which way is a fuse similar to a switch in a simple electric circuit?

19. Why should a nurse shake the clinical thermometer before it is used on another patient?

20. State one way in which a farmer can control crop diseases without using chemicals.

21. Name the turning point of machine.
-
22. Why ventilators are usually put on a higher level than doors and windows in a house?
-
23. How do conifers differ from other non-flowering plants?
-
24. What is the importance of chimney in kitchen?
-
25. Apart from getting food, give any one other way in which birds benefit from plants.
-
26. What type of skeleton does a cockroach have?
-
27. What is the importance of water during seed germination?
-
28. How does a human body cool itself on a hot day?
-
29. How is the fertilization of the eggs of amphibians different from those of reptiles?
-
30. How can the pitch of a string of a musical instrument be changed?
-
31. Mary shouted in a big house and she heard the same sound repeated. Name the repeated sound she heard.
-
32. Give one disadvantage of using herbicides (chemicals) to kill weeds in crop gardens.
-
33. State an example of a crop which is legume.
-
34. Give any one way in which the human liver can be kept in proper working condition.
-

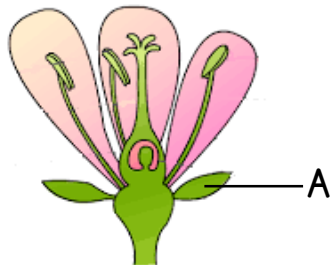
35. State any one way in which termites depend on soil.

36. A man lives with his wife, two sons and a nephew. What type of family is this?

37. Name any one by-product of respiration.

38. State the type of change that takes place when a baby grows to an adult.

The diagram below is a section of a flower. Use it to answer questions 39 and 40.



39. Name the part marked A.

40. Use letter P to show the part that produces the male reproductive cells.

SECTION B: 60 MARKS

41. a. Give one way in which Primary Health Care (PHC) is important to a community.

b. State the element of Primary Health care which is promoted by each of the following activities:

– Sweeping the kitchen

– Putting drops of polio vaccine into the mouth of a child

– Eating a balanced diet

42. The diagram below is of a fish. Use it to answer questions that follow.



a. Name the part labeled R.

b. Give the function of the parts labeled P and U.

P _____

U _____

c. What structure is covered by the part labeled Q?

43. a. State two differences between birds and mammals.

i. _____

ii. _____

b. Give two similarities between birds and mammals.

i. _____

ii. _____

44. The diagram below shows a germinating seed. Use it to answer questions that follow.



a. State the type of germination shown above.

b. Where does the seed above store the food used during germination?

c. Give an example of a crop with the same type of germination.

d. Apart from moisture, state any one other condition necessary for seed germination.

45. a. State any two natural causes of environmental degradation.

i. _____

ii. _____

b. Give two human causes of environmental degradation.

i. _____

ii. _____

46. a. Which part of the body is affected by each of the following diseases?

– Scabies _____

– Trachoma _____

b. State the vector which spreads each of the following diseases:

– Bilhazia _____

– Rabies _____

47. a. Mention two primary sex changes that happen in boys.

i. _____

ii. _____

b. Give two secondary sex changes that happen in both boys and girls.

i. _____

ii. _____

48. a. Apart from millet, wheat and sorghum, give any one other cereal food crops.

b. Mention any one common pest that destroys cereal crops.

c. Give any two reasons why cereal crops are dried before storing.

i. _____

ii. _____

49. a. Apart from a worker bee, mention two other types of bees found in a bee hive.

i. _____

ii. _____

b. Write down two materials collected by a worker bee from the environment.

i. _____

ii. _____

50. a. What form of energy does a ball have before it is kicked?

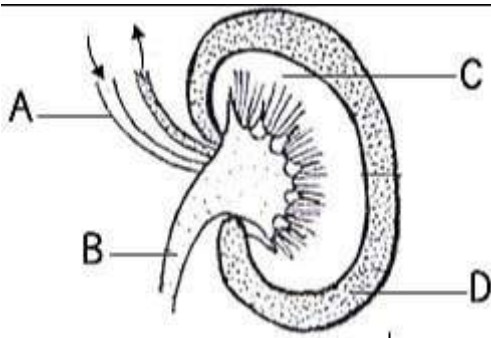
b. What energy change takes place immediately a ball is kicked?

c. Give two forms of energy produced by the ball as the goal keeper catches it.

i. _____

ii. _____

51. The diagram below is a section of a human kidney. Use it to answer questions that follow.



a. Name the parts marked with letters B and C.

B _____

C _____

b. What happens to blood in the part marked with letter D?

c. How is part A important to the kidneys?

52. a. Give any two reasons why it is important to wash hands before eating food.

i. _____

ii. _____

b. Apart from washing hands, mention any two other activities one can do to maintain personal hygiene.

i. _____

ii. _____

53. a. Name two natural resources in the environment that enable green plants to make their food.

i. _____

ii. _____

b. What is the name of the process in (a) above?

c. Which gas is given off during this process?

54. a. Name any two products got from sheep.

i. _____

ii. _____

b. What does shearing refer to in sheep rearing?

c. How can a farmer control pneumonia in sheep?

55. a. What force makes a pencil to reduce in length as one writes?

b. Name any one form of energy produced by the force in (a) above as one writes.

c. Give any two disadvantages of the force you have named above to the writer.

i. _____

ii. _____

SCIENCE PLE 2010

CANDIDATE'S INFORMATION

Index number :

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Name :

Signature :

School name :

District name :

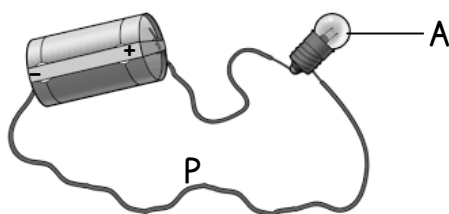
SECTION A: 40 MARKS

1. Name the sense of organ for hearing.

2. Give one item used for cleaning hands after visiting a latrine or toilet.

3. State one way in which rain may be a disadvantage to a crop farmer

The diagram below is of simple circuit. Use it to answer question 4 and 5.



4. Draw an arrow on line P to show the direction of the flow of electricity.

5. Apart from light, which other form of energy is produced at A?

6. What is the importance of dipping cattle?

7. Write down one bad behavior which is a result of drug abuse.

8. Name one type of seed dispersal.

9. Sunbirds visit flowers to get nectar. How do plants benefit from these birds?

10. Give one way in which the cause of a scald is similar to that of a burn.

11. Name the best soil for making pots.

12. Write down one block in a food path.

13. Give one way in which health education is important in controlling overpopulation.

14. How can farmer make use of cow dung produced by his animals?

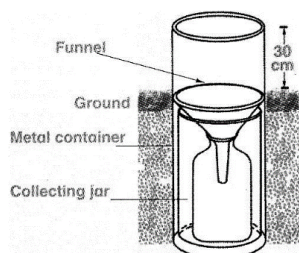
15. State one way in which hens are important in prevention of deficiency diseases in man.

16. Arrange the following organism in the correct order of a food chain.

Caterpillar → Dog → Leaves → Bird

17. Why is a person with hookworms likely to suffer from anemia?

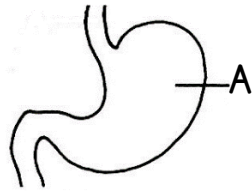
18. The diagram below is of weather instrument. Study it and answer question 18.



What is the use of the instrument in a weather station?

19. Give a reason for keeping *Ebola* victims away from other people.

The diagram below shows part of the human digestive system. Use it to answer question 20.



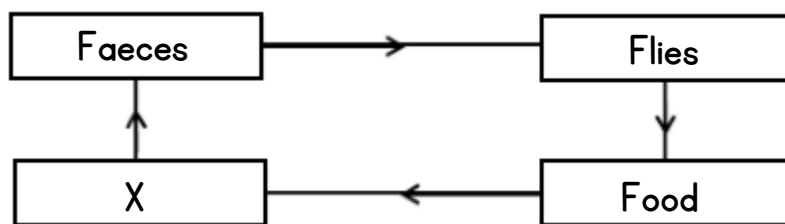
20. Name the class of food which is digested in the part marked A.
-
21. Write down one way in help in the control of cholera in a community.
-
22. Give one way in which the preservation of millet and fish differs.
-
-
23. State one way of controlling nagana in an area.
-
24. Give one example of a plant with prop roots.
-
25. Name the type of family where there are parents and their biological children only.
-
26. Write down one function of joints in human body.
-
27. You are a health prefect in your school. Give one activity you would do during a health parade.
-
28. How is wind useful to human beings?
-
29. Why is mercury used in a thermometer instead of water?
-
30. State one natural source of sound energy.
-
31. Write down the function of testes in a human reproductive system.
-

32. Give the difference between river blindness and night blindness.
-
-
33. How is thermos flask helpful in a home?
-
34. Apart from absorption of water and mineral salts, state one other use of roots to a cassava plant.
-
35. State one effect of heat on a metal.
-
36. How does mulching help to keep moisture in the soil?
-
37. What determine the size of the image in a pinhole camera?
-
38. Give one way in which energy from animals can be used.
-
39. Write down First Aid you would give to a person who has been bitten by a poisonous snake on the lower part of the leg.
-
40. Why does a coin sink when it is placed in a basin full of water?
-

SECTION B: 60 MARKS

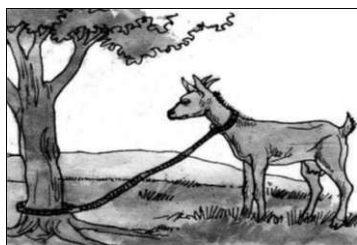
41. a. Name one natural and artificial source of light.
- i. natural _____
- ii. artificial _____
- b. Why is the moon not regarded as source of light?
-
- c. Give one way in which reflection of light is important to man.
-

42. The diagram below shows how diseases can spread through the 4Fs. Use it to answer question 42.



- a. Give the word which should be in the box marked X.
- _____
- b. Name two activities which can be done to avoid the spread of diseases through food.
- i. _____
- ii. _____
- c. What structure on the body of a fly makes it spread diseases?
- _____

43. The diagram below shows one of the methods of grazing goats in Uganda. Study it and use it to answer question 43.



- a. Name the method of grazing goats shown in the diagram.
- _____
- b. Give two advantages of using the method shown in the diagram.
- i. _____
- ii. _____
- c. Write down one disadvantage of using this method.
- _____

44. Match the item in list A with their functions in list B.

List A	List B
Red blood cells	stop bleeding when the skins is cut.
Arteries	carry oxygen around the body .
Valves	carry blood away from the heart.
Platelets	prevent back flow of blood in veins.

- a. Red blood cells _____
- b. Arteries _____
- c. Valves _____
- d. Platelets _____

45. a. Give two benefits of planting trees in your school compound.
- i. _____
 - ii. _____
- b. What two things can you do to protect in your school compound?
- i. _____
 - ii. _____

46. a. Name the type of cattle kept for milk production.
- _____
- b. Give one example of exotic breeds of cattle which are the best for milk production.
- _____
- c. Write down two activities a farmer should do in order to get clean milk from a cow.
- i. _____
 - ii. _____

47. a. State one Primary Health Care activity children can do to help their mothers when cooking.
- _____
- b. Write down three health care activities that can be provided to the elderly in our homes.
- i. _____
 - ii. _____
 - iii. _____

48. a. Name two scavengers in the environment.
- i. _____
 - ii. _____
- b. In which way are scavengers useful in the environment?
- _____
- c. Give the difference between a predator and scavenger.
- _____
- _____

49. a. Which organ removes urine from the body?

b. To which body system does the organ that removes urine from the body belong?

c. Where does urine collect before it is removed from the body?

d. Name one substance found in urine.

50. a. Name two water resources in Uganda that provide us with proteins.

i. _____

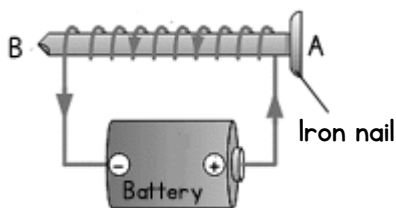
ii. _____

b. Give two water human practices that can lead to the destruction of such resources.

i. _____

ii. _____

51. The diagram below shows a method of making a magnet. Study it and use it to answer question 51.



a. Name the method of making a magnet shown in the diagram.

b. What pole will the part marked A become?

c. What would you do to the dry cells in order to increase the strength of the magnet?

d. Apart from the method shown in the diagram, name any other method of making a magnet.

52. a. Name the germ that causes HIV/AIDS.

b. Apart from having sex an infected person, write down two other ways in which one can get the germ that causes HIV/AIDS rural areas.

i. _____

ii. _____

c. ABC is a way of controlling the spread of the germ that causes HI/AIDS. What does A stand for?

53. a. Apart from killing people, give two other dangers of lightening.

i. _____

ii. _____

b. Write down two ways by which people can protect themselves from the dangers of lightening.

i. _____

ii. _____

54. In an experiment, salt is mixed with water and stirred until it dissolves to make a solution.

a. What do the following act as in the experiment?

- Salt _____

- Water _____

b. Name any other substance that could be used instead of salt.

c. How can salt be recovered from the solution?

55. a. Why are vaccines used in the prevention of certain diseases?

b. How is the measles vaccine given to a baby?

c. Apart from immunization, state two other ways in which a baby's immunity can be improved.

i. _____

ii. _____

SCIENCE PLE 2009

CANDIDATE'S INFORMATION

Index number :

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Name :

Signature :

School name :

District name :

SECTION A: 40 MARKS

1. Name of domestic animal used for protection in a home.

2. Give the function of a micropyle to germinating seed.

3. What general name is given to animals with a backbone?

4. Name the type of preservation where slow burning fire is used to dry food.

5. Give one pest which destroys the roots of root crops in gardens.

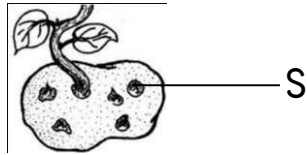
6. What state of matter is smoke?

7. Give a reason why a tsetse fly is dangerous to cattle.

8. How is reproduction in beans different from that in ferns?

9. Write one activity that can help to reduce snake bites in homes

10. The diagram below is of an Irish potato tuber. Use it to answer the question that follows.



What is the use of the part marked S?

11. Why is immunization of children carried out free of charge in Uganda?

12. Write one reason why people working in dirty areas should wear gumboots.

13. Why should young girls avoid getting pregnant?

14. What type of farmer use in order to harvest both crops and woods?

15. The diagram below shows a method of making water clean. Use it to answer question that follows:



Name the method used above.

16. Why do objects thrown in the air fall back on the ground?

17. Why should communities fence open spring wells?

18. State one way of preventing athlete's foot disease.

19. Why is an ostrich unable to fly?

20. Apart from trachoma, name one other eye disease.

21. State one way holes in the soil created during brick-making may contribute to the spread of malaria in an area.
-
22. State one natural cause of a family failing to have enough food.
-
23. Name one material used to guard against electric shocks.
-
24. Give one activity the community can do to reduce road accidents.
-
25. Name one activity carried out by farmers to remove unwanted plants from their gardens.
-
26. Give one way in which claws are important to birds.
-
27. To which group of simple machines does a ladder leaning against a wall belong?
-
28. Why is it a bad practice to store cooked food in a room where domestic animals are kept?
-
29. Write one method of rearing cattle used in towns.
-
30. State one similarity in structure between a snails and an earthworm.
-
31. A part from rock particles and living organisms, name one other part of soil.
-
32. What happens when an opaque object is placed in a path of light?
-
33. Name the kind of energy possessed by water at 100°C .
-
34. Give one factor to be considered when choosing a site for a school garden.
-

35. The diagram below shows a stage of development of an animal. Use it to answer the question that follows.



Name one animal that develops from this stage.

36. State one common problem which adolescents face.

37. Why is breast feeding very important to a mother?

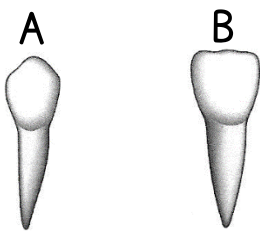
38. How do plants reduce the amount of carbon dioxide in the atmosphere?

39. Name the part of a flower that holds the anther ahead.

40. Give one reason why people who get badly burnt are given plenty of fluids.

SECTION B: 60 MARKS

41. The following diagram shows two types of teeth in human beings. Use it to answer questions that follow.



- a. Name the type of tooth marked A and B.

A _____

B _____

- b. Give the use of each of the teeth in the diagram above.

A _____

B _____

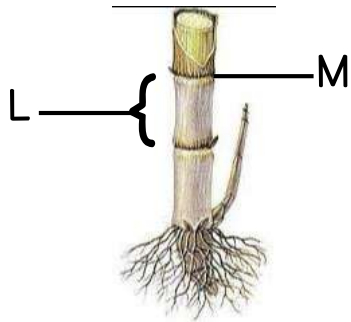
42. a. How does a guitar produce sound when its string is plucked?

- b. How can the pitch of a guitar be changed?

- c. Name one way of storing sound

d. Give reason why sound cannot travel through a vacuum.

43. The diagram below is of a sugarcane plant. Use it to answer the question that follows.



a. Name the parts of marked with letters M and L.

M _____

L _____

b. What food value is got from eating sugarcane?

c. How is such a sugarcane plant propagated?

44. a. State one way in which nursery bed is important to a farmer.

b. Name two vegetable crops commonly grown in a nursery bed.

i. _____

ii. _____

c. Give one way in which farmer can care for crops in a nursery bed.

45. a. Write one dangerous substance found in tobacco smoke.

b. Give the meaning of each of the following:

– Active smoking

– Passive smoking

c. What piece of advice can you give to smokers to help them to stop the habit?

46. The density of sand is 3 grams per cubic centimeter. Its mass is 270 grams. Find its volume.

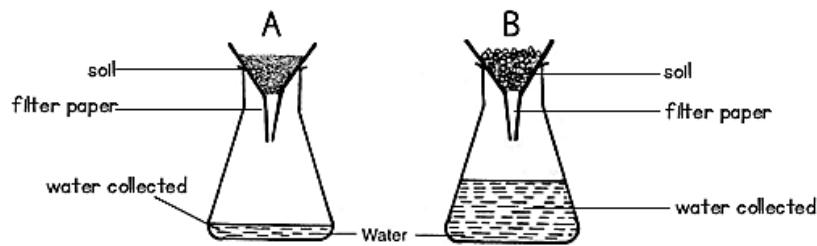
47. a. What is First Aid?

b. List any two things found in First aid box.

- i. _____
- ii. _____

c. State why every school should have a First Aid box.

48. Equal amounts of water were produced onto soil A and B shown in the diagram below:



a. From which soil did more water drop?

b. State why more water dropped from soil you have named in (a) above.

c. Name the type of soil in B.

d. Give one use of the type of soil in A.

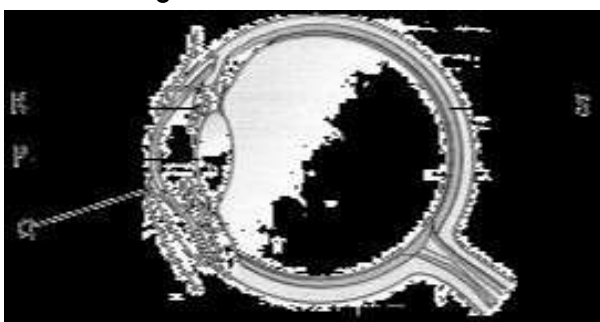
49. a. Write two activities that help to promote good sanitation in a home.

- i. _____
- ii. _____

b. Name two diseases that may attack a family due to poor sanitation.

- i. _____
- ii. _____

50. The diagram below shows a human eye. Use it to answer the question that follows.



a. Name the parts marked P and Q.

P _____

Q _____

b. What is the function of part marked R?

c. Which letter on the diagram shows where images are formed?

51. The table below shows weather instruments, and what they measure. Match the items in list A with those in list B.

List A	List B
Wind vane	amount of rainfall.
Barometer	speed of wind.
Rain gauge	amount of air pressure.
Anemometer	direction of wind.

a. Wind vane _____

b. Barometer _____

c. Rain gauge _____

d. Anemometer _____

52. a. What name is given to the process that turns sugar to alcohol?

b. Apart from making alcohol, state two other ways the process you have named in (a) can be useful to human beings.

i. _____

ii. _____

c. Name the fungus used in this process.

53. a. Name the process that takes place in the air sacs of a human lung.

b. What prevents the trachea (wind pipe) from closing during breathing?

c. Give two products found in the air we breathe out.

i. _____

ii. _____

54. a. State the function of each of the following structures in the human skeleton.

- Ligament _____

- Tendon _____

b. Give one example of a long bone in a human skeleton.

c. Name one disease that affects a human skeleton.

55. a. Name the germ that causes gonorrhea.

b. Give one sign of gonorrhea infection in a newly born baby.

c. How does gonorrhea infection increase chances of a person getting HIV/AIDS?

d. How can a person avoid getting gonorrhea?

SCIENCE PLE 2008

CANDIDATE'S INFORMATION

Index number :

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Name :

Signature :

School name :

District name :

SECTION A: 40 MARKS

1. Name one body organ in man that makes use of light energy from the sun.

2. Name one cold-blooded vertebrate that has a body shell.

3. Mention any one way in which a biological change is similar to a chemical change.

4. How can a passion fruit plant benefit from growing near a mango tree?

5. Write down the vector that spreads elephantiasis.

6. To which group of animals does the one shown in the diagram below belong?



7. Give any one use of wearing sun grass.

8. Why is a maize grain grouped under monocotyledonous seeds?

9. What is the importance of a web to a spider?

10. How does fish help in controlling the spread of malaria?
-
11. Mention any one way in which communicable diseases are spread.
-
12. Whenever peter opens or closes his door, the hinge makes noise. What can he do to stop the noise when he is opening?
-
13. Name one structure in a human body which connects bone to bone.
-
14. Give any one way in which petrol in a car is similar to food in a human body.
-
15. How are ferns, mosses and liverworts similar?
-
16. State any one health reason why the building of industries should be limited in an area?
-
17. Why should you chew food before swallowing it?
-
18. How is a chameleon important to human beings?
-
19. Suggest any one use of a tourniquet.
-
20. State any one way in which soil is important to insects.
-
21. What system is responsible for the removal of waste materials from our bodies?
-
22. State one result or two of naked wires carrying electric current getting in touch with one another.
-
23. Give the function of the epiglottis in the respiratory system.
-

24. Write down the type of venation shown by the diagram below.



25. State the organisms that can be controlled by use of antibodies.

26. Write down any one piece of advice you can give to a friend in order to promote the person hygiene.

27. Why is a microscope important to doctors?

28. How does boiling of water help the control of typhoid?

29. State the function of the amniotic fluid during pregnancy.

30. Which type of electricity is produced by generators?

31. Give any one cause of malnutrition in adults.

32. How is TASO (The Aids Support Organization) useful to HIV/AIDS victims?

33. What process makes dew to disappear during hot day?

34. Give any one use of reptile to man.

35. How are green plants important in the water cycle?

36. Mention any one natural resource found under the ground and is used as fuel.

37. Apart from getting medical treatment, write any other piece of advice you can give to a friend suffering from flu.

38. Name the state of matter in which sound travels fastest.
-
39. Why is it easier for Otim to push a wheel barrow full of manure downhill than uphill?
-
40. State an immisable disease which enters our bodies through cuts and wounds.
-

SECTION B: 60 MARKS

41. Use the diagram below to answer questions that follow.



- a. What name is given to the above equipment?

- b. Why is the above equipment always painted with bright colors?

- c. Why is such an equipment recommended to be in schools?

- d. What gas is contained in the above equipment?

42. a. A part from the skin, give any one example of a respiratory organ.

- b. List any two diseases which affect the respiratory organ in humans.

i. _____

ii. _____

- c. Give any one practice by humans which may lead to a respiratory disease.

43. a. Suggest two reasons why young farmers' clubs are important in schools.

i. _____

ii. _____

- b. Give any two activities that can be done by young farmers' clubs in schools.

i. _____

ii. _____

44. a. Name any two of the 5RS in waste management.

i. _____

ii. _____

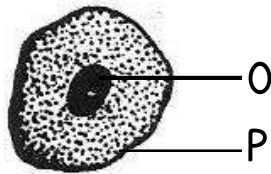
b. Suggest any two effects of poor waste management in the environment.

i. _____

ii. _____

45. The diagram below is of an amoeba. Use it to answer questions that follow.

a. What kind of organism is an amoeba?



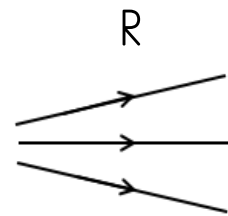
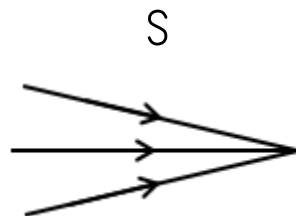
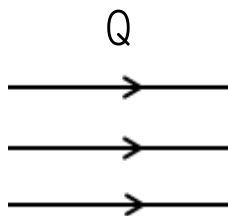
b. Name the structure indicated by letters.

O _____

P _____

c. How is the amoeba able to move from place to place?

46. The illustrations below show different types of light rays. Use them to answer questions that follow.



a. Name the following rays of light.

Q _____

S _____

b. Suggest the type of lens that can be used to make light rays move as shown in S above.

c. What eye defect is corrected by a lens which makes rays move as shown in R?

47. A poultry farmer has his chicken passing out blood stained droppings and have drooping wings.

a. What disease is likely to have attacked the farmer's birds?

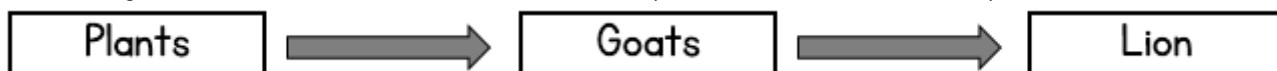
b. Suggest two ways in which the neighbors to the farmer can prevent their birds from contracting the above disease.

i. _____

ii. _____

c. Why should the farmer separate the birds with the above given conditions from the rest?

48. The diagram below is of a food chain. Study it and answer the questions that follow.



a. Name the producer.

b. Which living organism in the diagram is herbivorous?

c. Why is the organism in (b) above herbivorous?

d. Name one way in which plants benefit from the goats.

49. The diagram below is of a pulley system. Use it to answer questions that follow.



a. Name the type of pulley system shown above.

b. Use an arrow to show the direction of effort.

c. If the mechanical advantage of the machine is 2 and the load being carried by the machine 50 kg, find the effort needed to raise that load.

50. a. Give two reasons for placing utensils on a rack.

i. _____

ii. _____

b. Why should a rack be constructed with stands at least one meter above the ground?

c. Suggest one danger of leaving utensils for a long time outside on a rack.

51. a. Mention any one method of harvesting root crops.

b. Give any three ways in which weeding helps on the proper growth of root crops.

i. _____

ii. _____

iii. _____

52. a. What name is given to the liquid part of blood?

b. Name one component of the liquid part of blood.

d. Give any two uses of the liquid part of blood.

i. _____

ii. _____

53. Give the meaning of the following words as used in cattle-rearing.

a. Insemination:

b. Cross-breeding:

c. Castration:

d. On heat:

54. a. What is the meant by the term silting?

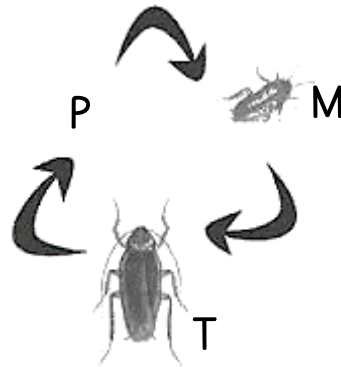
b. Apart from killing fish, how else does silting affect fish?

c. List any two major causes of silting.

i. _____

ii. _____

55. The diagram below shows a life cycle of an insect. Use it to answer the questions that follow.



a. Name stages marked M and T.

M _____

T _____

b. Name one insect that undergoes the above stages of development.

c. Draw the stage marked with letter P.

SCIENCE PLE 2007

CANDIDATE'S INFORMATION

Index number :

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Name :

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SECTION A: 40 MARKS

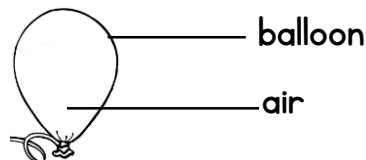
1. Which reptile changes the color of its body it moves from one place to another?

2. Give any one way in which bush burning leads to soil erosion.

3. How are bats different from hens in their way of reproduction?

4. How does immunization promote health in a child?

5. Give any one property of air shown in the diagram below.



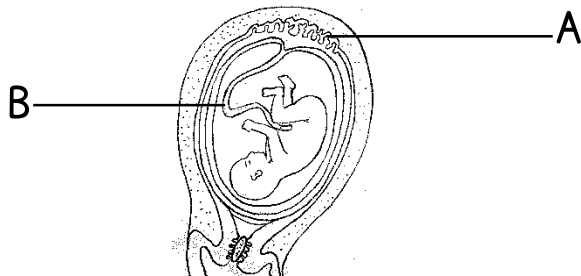
6. Give the difference between evaporation and condensation.

7. Why is an earth-worm called an invertebrate?

8. Give any one function of a human skeleton.

9. Give any one way in which the spread of scabies can be controlled.
-
10. Mention a disease which makes a person to pass out watery stool with blood.
-
11. State the difference between volume and mass.
-
-
12. Give any one body characteristic which is common to a spider and a tick.
-
13. Which is the main cause of fainting?
-
14. Suggest any one way in which human beings use energy from the sun?
-
15. Give any one way in which sweating is important in a human body.
-

The diagram below shows a fetus in the womb. Use it to answer questions 16 and 17.



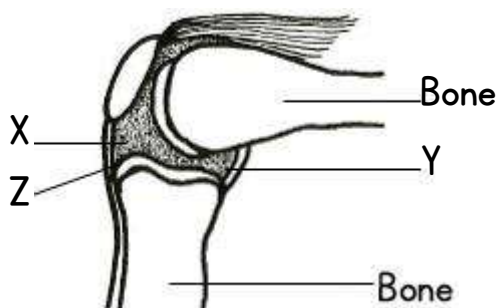
16. Name the part marked with letter A.
-
17. Name any one way in which the part marked with letter B is useful to the fetus.
-
18. Which part of the human eye works like a camera?
-
19. Give any one activity that can be done at home to control the spread of malaria.
-
20. In which way is a person who is HIV positive different from the one who has developed AIDS?
-

21. Give any one way in which a transparent object differs from an opaque object.
-
22. What causes kwashiorkor in children?
-
23. Why is photosynthesis important to plant?
-
24. Give any one reason why human beings breathe in.
-
25. What is an enzyme?
-
26. Give any one reason why plants and animals are classified as living thing.
-
27. State any one way in which health parades are important in primary schools.
-
28. Apart from painting, give any one other way in which metals can be prevented from rusting.
-
29. Which type of soil does not allow water to go through it easily?
-
30. Why is a sling used when giving First Aid to a person with a broken hand?
-
31. State any one practice on a school garden that will help to improve soil fertility.
-
32. Why is a tick called a parasite?
-
33. State the difference between the function of a pulmonary vein and that of a pulmonary artery in blood circulation.
-
34. Give any one form of energy that is produced during the process of splitting firewood with an axe.
-

35. Give any one way in which plants benefit from animals.
-
36. Give any one sign of a school compound which shows that the school environment is degraded.
-
37. In which way the function of the kidneys similar to that of the skin in a human body?
-
38. A part from the brain, give any one other body organ which is greatly damaged by alcoholism.
-
39. How does heat from the sun reach the ground?
-
40. Give any one way in which teeth are important in the digestion of food.
-

SECTION B: 60 MARKS

41. The diagram below shows a human joint. Use it to answer the questions that follow:



- a. Name the parts marked with letters Y and Z

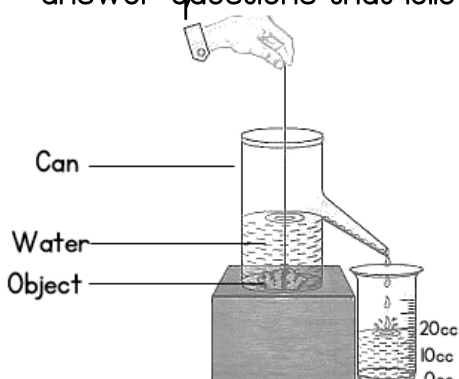
Y _____

Z _____

- b. Give any one function of the fluid found in the place marked with letter X.

- c. What type of joint is shown in the diagram above?

42. An experiment was done by P7 class as shown in the diagram below. Use it to answer questions that follow.



- a. When is such a method of finding volume used?

- b. If the object above has a density of 2 g/cm^3 , find its mass.

43. a. Give any two characteristics that are common to a frog and a crocodile apart from being vertebrates.

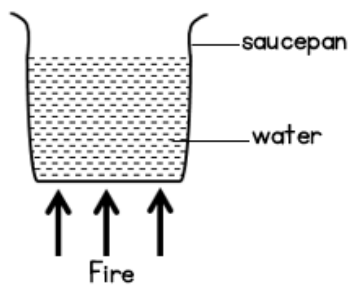
i. _____

ii. _____

b. How is fertilization of a frog different from that of a crocodile?

c. Name any one other vertebrate that can be classified with a frog.

44. The diagram below shows a saucepan with some water on fire. Using it to answer the questions that follow:



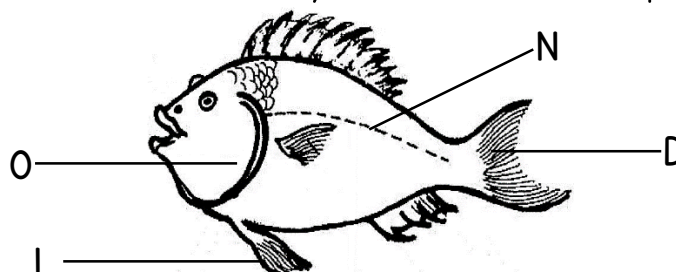
a. How will heat from the fire move to reach the water at the bottom of the saucepan?

b. Which method will enable the heat to reach all the water in the saucepan?

c. Use arrows to show the movement of water in the saucepan after it has been heated for some time.

d. What will happen to the water in the saucepan to show that it is boiling?

45. The diagram below is of a fish. Study it and answer the questions that follow:



a. Name the structure marked L.

b. What is the function of each of the structure marked D and N?

D _____

N _____

c. State the function of the structure covered by the part marked O.

46. The diagram below shows a seedling in a box. Study it and answer the questions that follow.



a. What color will the leaves of the seedling be if left in this box with H covered for a week?

b. Give a reason for your answer in (a) above.

c. Give any two things are likely to happen to the seedling if the part marked with letter H is left open.

- i. _____
- ii. _____

47. a. Give any one reason why a magnet can not attract pieces of wood.
-

b. Give any three practices that can lead to the destruction of a magnet.

- i. _____
- ii. _____
- iii. _____

48. Kamoga is a poultry farmer. He has 100 layers on his farm. Give any four reasons why he should keep farm records.

- i. _____
- ii. _____
- iii. _____
- iv. _____

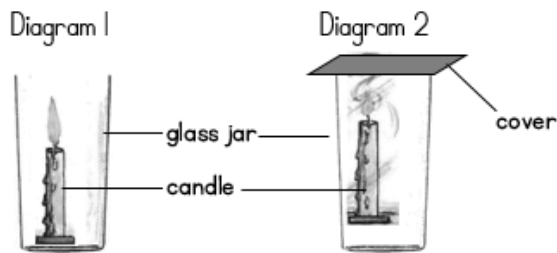
49. a. Why is it a bad practice to apply soil or cow dung on any burnt area of our skin?
-

b. Suggest any two reasons why the injured part with a burn or scald is dipped into cold water.

- i. _____
- ii. _____

c. Give any one way of avoiding burns while lifting hot objects.

50. The diagram numbered 1 and 2 below show an experiment done by a P.7 class. Diagram 1 show a burning candle in an open jar. In diagram 2, the jar was covered as shown. Use the diagram to answer the questions that follow:



a. What would happen to the flame in diagram 2 if the cover was removed after a short time?

b. Give a reason for your answer in question (a) above.

c. If the cover have been left on the jar in diagram 2 for a long time, what would have happened to the flame?

d. Give a reason for your answer in question (c) a.

51. a. What is environmental degradation?

b. Give any three natural causes of environmental degradation.

- i. _____
- ii. _____
- iii. _____

52. a. Give any two ways in which mulching is important to the soil.

- i. _____
- ii. _____

b. Suggest any two materials that are commonly used for mulching gardens.

- i. _____
- ii. _____

53. a. State any one function of each of the following parts in a simple electric circuit:

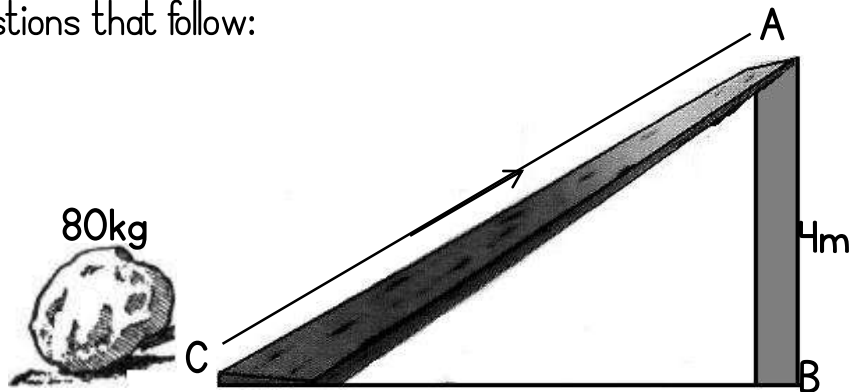
- Wire _____
- Battery _____

b. Mention any two causes of a short circuit.

i. _____

ii. _____

54. The diagram below shows a machine used for lifting loads. Use it to answer the questions that follow:



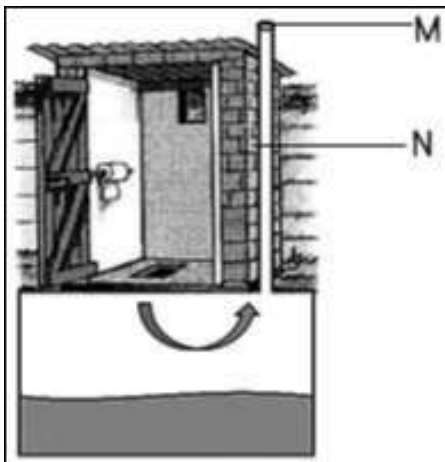
a. What distance is the load to be lifted?

b. Give the advantage of lifting the load alongside CA.

c. Find the mechanical advantage of the machine above.

d. How can the machine be improved so that less effort is used to lift loads?

55. The diagram below is of a Ventilated improved Pit (VIP) latrine. Use it to answer the questions that follow:



a. Name the part marked M and N.

M _____

N _____

b. Give the function of the part marked M.

c. What does the arrow in the diagram show?

SCIENCE PLE 2006

CANDIDATE'S INFORMATION

Index number :

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Name :

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SECTION A: 40 MARKS

1. What good health habit should be practiced after visiting a latrine or toilet?

2. Give any one way in which animals are important during the process of photosynthesis.

3. What form of energy does falling stone have before it reaches the ground?

4. State any one reason why some plants climb others?

5. Give one common use of sand at home

The diagram below shows a glass of water with water and a pencil in it. Use it to answer question 6.



6. What causes the pencil to appear bent?

7. What first Aid would be given to Jamada who has burnt his hands with hot metal?

8. Give any one practice that can help to control the spread of cholera in a school.

9. Give any one reason why people plant trees around their houses.

10. A part from the skin and lungs, name any other excretory organ.

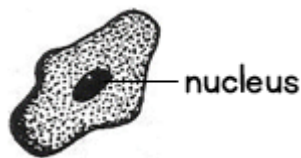
11. Give any one way in which the breathing of a housefly is different from that of a rat.

12. Give any one way in which soil is important to a plant.

13. Why is a wet cloth put on the body of a person with fever?

The diagram below is of a blood cell. Use it to answer questions 14 and 15.

14. What type of blood cell is shown in the diagram?



15. Give a reason for your answer in question 14 above.

16. Give any one reason why plants are grouped among living things.

17. Give one example of a leguminous crop

18. How does a snail protect itself from its enemies?

19. Give any one disadvantage of keeping poultry on free range system.

20. Give any one reason why air is considered to be matter.

21. Give any one reason why people like the brick makers need to feed on more carbohydrates.

22. Which part of air is necessary for burning to take place?

 23. Give the meaning of drug abuse.

 24. State one way of making a magnet.

 25. Which part of the human body does trachoma attack?

 26. Suggest any one reason in which the use of solar energy helps in the protection of the environment.

 27. Why is a ventilated improved Pit latrine (VIP) left without a cover?

 28. How do some people become passive smokers?

 29. Draw a prop root system in the space provided below.
-
30. Name any one crop with a root system similar to the one you have drawn above.

 31. What causes an echo?

 32. Why does a stone sink when it is put in a bucketful of water?

 33. Give one way through which worms enter our bodies.

 34. Mention the instrument used to measure the temperature of a human body.

35. What is the process by which seeds grow into seedlings?

36. State any one use of an axe in a home.

37. Give any one reason why white objects are seen easily even at a distance.

38. Give any one way in which people benefit from bees.

39. Which bone in the human skeleton protects the brain?

40. Give any one characteristic which is common in all the invertebrates.

SECTION B: 60 MARKS

41. John fell off a bicycle and broke his high bone.

a. Name the injury which John got.

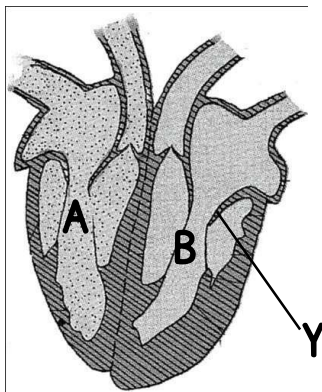
b. State two ways in which John can be given first aid.

i. _____

ii. _____

c. What is the importance of giving first Aid to a person like John?

42. The diagram below shows a human heart. Use it to answer the questions that follow:



a. What is the function of the part marked letter Y?

b. Give the difference between the blood in the regions marked with letters A and B.

c. Put arrows on the short lines marked with number 1 and 2 to show the direction of the movement of blood.

43. a. Name any two parts of a plant which provide human beings with food.

i. _____

ii. _____

b. Give any two ways in which plants benefit from animals.

i. _____

ii. _____

44. Below is a list of activities carried out by people in the environment. Use the list to answer the questions that follow.

- Fishing
- Producing a lot of smoke in the factories
- bush burning
- Draining swamps

a. Which activity is likely to decrease the amount of rainfall?

b. Which activity is likely to spoil the fertility of the soil?

c. Which activity is good for controlling crop diseases?

d. Which activity is likely to cause pollution of the atmosphere?

45. The table below shows different groups of vertebrates.

A	B	C
Birds	Mammals	Amphibians

a. To which group of vertebrates does each of the following belong?

- Bat _____

- Frog _____

b. Give any one way in which the reproduction of animals in group B is different from that of animals in group C.

c. Give any one other group of vertebrates which is missing in the table.

46. a. What is the importance of taking children for immunization?

b. Which vaccine is given to the child in the mouth at birth?

c. Give any two ways in which the child's health is important to the parents.

i. _____

ii. _____

47. a. Give any two reasons why a mushroom is not a plant.

i. _____

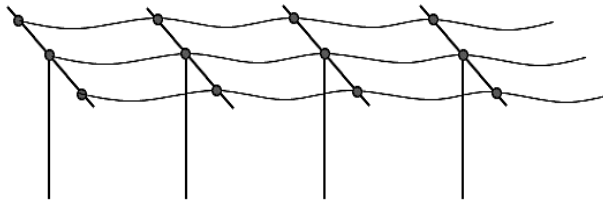
ii. _____

b. Give any two plants that produce in the same way like a mushroom.

i. _____

ii. _____

48. The diagram below shows wooden poles with electricity wires. Use it to answer the questions that follow:



a. State any one reason why the wires are loosely fixed.

b. What would happen to wires if they were tightly fixed?

c. Give one reason why wooden poles are usually used to carry electricity.

d. Why are wires placed very high up?

49. a. Give any one part of the human body where each of the following joints is found.

– Ball and socket joint

– Hinge joint

– Gliding joint

b. Which one of the above joints can make an all-round movement?

50. a. What is a chemical change?

b. Give any one example of a chemical change.

c. When water changes to ice, the process is called freezing. Name the process in each of the following changes.

– Ice to water _____

– Steam to water _____

51. a. Give any one reason why people include proteins in the diet.

b. What is the importance of fruits in a diet?

c. State any two ways of protecting children against malnutrition.

i. _____

ii. _____

52. The list below gives some of the methods farmers use to control soil erosion. Use it to answer questions that follow.

– Terracing

– planting trees

– Contour ploughing

– Mulching

Which of the two methods given above are best for?

Land with a gentle slope

i. _____

ii. _____

Land with a steep slope

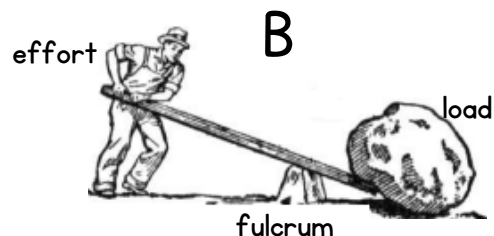
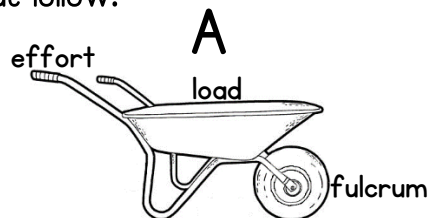
i. _____

ii. _____

53. Give any one sign you would see on cattle suffering from the following diseases.

- a. Anthrax _____
- b. East Coast fever _____
- c. Nagana _____
- d. Foot and mouth _____

54. The diagram below shows two simple machines. Use it to answer the questions that follow:



a. Name the machine marked with letter A

b. State the class of levers to which each of the above machines belongs.

- Machine A _____
- Machine B _____

c. Give any one example of a machine which works in the same way as the machine marked with letter B.

55. The diagram below shows a rabbit house. Use it to answer the question that follow:



a. Give any one reason why the rabbit house is raised above the ground.

b. Give any two reasons why wire mesh is used in building a rabbit house.

- i. _____
- ii. _____

c. State any one thing that should be provided to the rabbit in the house.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Blank lined paper for writing.